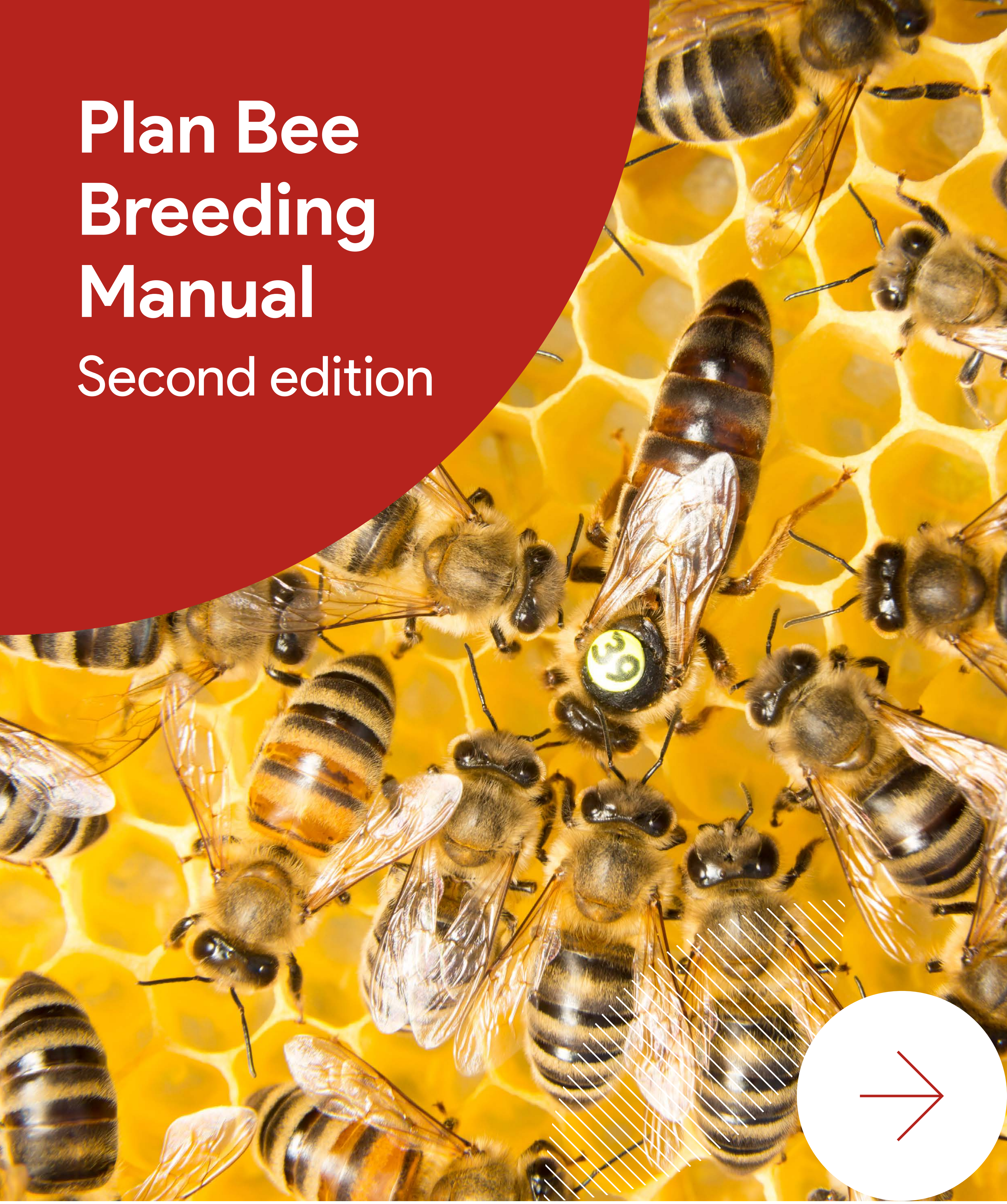
Plan Bee Breeding Manual

Second edition



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AgriFutures® Plan Bee

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Photo: Emily Noordyke

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Foreword

Many livestock industries have been revolutionised by modern animal breeding techniques, however the Australian beekeeping industry has been slower to adopt. Traditionally, bee breeding in Australia has involved keeping lines of bees, with breeders replacing the mother queen with her best daughter as required.

Honey bees are of critical importance to Australian agriculture, however new and emerging challenges and threats have necessitated greater focus on implementing modern animal breeding in the industry. Modern breeding uses standardised selection criteria such that animals owned by different breeders, or that are in different environments, can have their performance compared. Quantifying the genetic merit of individuals and using the best performers to breed the next generation can dramatically increase production and reproductive traits.

The potential for modern animal breeding to transform Australia's honey bee industry was the impetus for the national honey bee genetic improvement program, known as Plan Bee, a collaborative research program involving researchers, beekeepers, industry bodies, government and key stakeholders. From 2020 to 2024, Plan Bee focused on identifying and selecting traits of importance to beekeepers and horticulture and broadacre cropping, developing a national database to assist beekeepers with choosing breeding stock according to key traits.

This *Plan Bee Breeding Manual: Second edition* has been developed to help beekeepers and breeders implement modern breeding in their enterprises and improve the genetics of Australia's honey bee population.

The manual outlines the principles of modern breeding; provides guidance on recording performance data using a standardised national approach and submitting that data to the Plan Bee database; and contains easy-to-use recording sheets and checklists. A trait dictionary has been included to remove ambiguity when recording data.

Through engaging in the national genetic improvement effort, beekeepers and breeders can ensure the health and productivity of future generations of honey bees, and assure the viability of the industry amid an increasingly challenging operating environment. The impacts of a successful genetic improvement program are also experienced by the growers of horticultural and broadacre crops reliant on honey bees for pollination.

Lechelle van Breda
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This manual is designed for bee breeders – those who produce queens either for commercial sale or their own business. The first section outlines the theory behind modern animal breeding. The second section covers how to implement modern animal breeding in your business. The third section contains the trait dictionary.

Contents

Acknowledgements

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Photo: Casey Cooper

Glossary

BBCoP	Bee Biosecurity Code of Practice.
Bee breeder	A beekeeper who produces queens that have been selected for traits in a defined breeding program and who controls mating either through artificial insemination or isolated mating, selecting both the queens and drones.
Best linear unbiased prediction	The statistical method used to calculate estimated breeding values.
Box	Essential equipment to house a colony of bees in a hive; the box contains frames of comb. A box may be used either as a brood box or a super.
Breeder queen	A superior queen chosen to produce the next generation, either queens (queen mother) or drones (drone mother).
Breeding objective	The goal of the breeding program – the traits to be improved and the desired direction of selection.
Brood box	A box of frames that the queen is confined to such that bee brood will be found only in this box.
Colony	The family of bees within a hive.
Colony ID	An identification assigned in the Plan Bee database based on the hive ID, queen ID and when the queen was introduced.
Contemporary group	A group of colonies that have been managed under the same conditions.
Drone	A male bee.
Drone mother	A breeder queen chosen to produce drones.
Estimated breeding value	An estimate of the genetic merit of an animal relative to the population average.
Genetic merit	The value of the genes of the queen for the traits(s) of interest.
Genetic testing	Testing of DNA at genetic markers to construct genotypic data; also referred to as genotyping.
Genotype	All the genes in an organism.
Genotypic data	A subset of the genotype, as read via DNA markers collected through genetic testing. This is often referred to as the genotype of the organism, but does not reflect all genes in the organism.
Heritability	The proportion of variance that is due to genetics rather than the environment or other systematic effects (e.g. the age of the queen). The heritability of a trait indicates how much of the observed superiority of a particular queen will be passed on to her daughters.
Hive	The structure that houses a colony of bees.
Hive ID	The label on the hive as assigned by the beekeeper.
Load	A collection of beehives that are transported together and managed together in the same apiary.
Mating nuc	A small hive into which queen cells are introduced and in which virgin queens emerge and reside while they mate and commence egg laying; also referred to as a mating nucleus.
Measurement accuracy	How accurately a trait can be measured.
Pedigree	The record of parentage through generations. For bees, usually only the mother is known, but genotyping can establish more detailed pedigree.
Phenotype	The observable or measurable traits of an organism.
Production queen	Queen for general use by beekeepers.
Qualitative measure	A descriptive measure of a trait.
Quantitative measure	The exact quantity measure of a trait.
Queen	The sole reproductive female in a colony, producing drones, workers and queens. Under some circumstances, a worker may produce drones.
Queen excluder	A barrier placed between the brood box and the super to confine the queen to the brood box; the holes are big enough for workers to pass through, but too small for queens and drones to fit through.
Queen ID	A unique identification assigned to each queen in the database, consisting of information on user ID, queen age and a generic code.
Queen mother	A breeder queen chosen to produce queens.
Queen producer	A beekeeper who produces queens but does not perform selection on them through recording of traits. Typically, a queen producer does not control mating. Often, they produce queens from stock that someone else has bred.
Queen tag	A coloured and numbered plastic disk that can be stuck onto a queen's thorax.
Repeatability	How consistently a trait is valued when multiple measurements are taken across time. A trait is repeatable if it ranks similarly each time you evaluate a colony.
Super	A box of frames placed above a queen excluder for honey production.
Worker	A female bee that usually does not reproduce.

Introduction

Modern animal breeding techniques have been used in the dairy, beef, lamb, wool and pork industries for decades. In all cases, this has led to dramatic increases in production and reproductive traits (Ibtisham *et al.* 2017; Banks *et al.* 2020), and is essential for creating effective genetic change in any hard-to-measure, late-in-life or sex-limited (a trait expressed in only one sex) trait.

Traditional bee breeding in Australia has typically involved keeping lines of bees, with the best daughter queen replacing the mother queen as the breeder. Bee breeders may keep records on their hives, or they may simply place a mark on a hive to note that it is either particularly good or particularly bad. The best hives are then used to either produce queens, drones or both.

By comparison, modern breeding uses standardised selection criteria such that animals owned by different breeders, or that are in different environments, can have their performance compared. Implementation of modern bee breeding requires good recordkeeping not only of performance but of pedigree, and also to identify which hives have been kept together. Such recording enables environmental effects to be separated from genetic effects, and helps identify management that may advantage or disadvantage only some hives that are kept together. Rather than replacing a queen with her best daughter, modern breeding involves selecting the top performers, regardless of their maternity; lines are generally not kept. Modern animal breeding has been so effective because the genetic merit of individuals can be more accurately quantified for selection. Modern breeding techniques can also involve genetic data (i.e. going beyond pedigree), appropriate weighting or emphasis on different economically important traits, and tools to control inbreeding.

The superiority of modern animal breeding in the honey bee industry is demonstrated by the success of the *Apis mellifera carnica* (Carniolan honey bee) breeding program in Germany and surrounding countries. While using traditional line breeding techniques from 1970-1989, the program achieved on average an 0.11% increase in honey production and 0.01% improvement in temperament score per year. Modern animal breeding techniques were subsequently introduced in the 1990s but were little used in breeding decisions until about 2000 (Hoppe *et al.* 2020). Selection of replacement queens was based on paternal descent, direct and maternal effects, and multi-trait models, and there was a shift away from line breeding during this period (Hoppe *et al.* 2020). From 2014-2018, the program achieved a 1.67% increase in honey production and a 1.45% improvement in temperament score (Hoppe *et al.* 2020).

What does that mean for you? If your average hive produced 55 kg of honey per year at the start of a breeding program, using traditional techniques you could expect to produce 0.0605 kg extra honey each year, or 0.9075 kg extra over 15 years. For comparison, the use of modern techniques would allow you to produce 0.9185 kg extra honey each year, and 13.7775 kg over 15 years. Assuming a wholesale honey price of $4.50/kg, the traditional program would increase income by $4.08 per hive after 15 years, while the modern program would result in $62 more per hive after the same period. We do not know what the rate of improvement will be in Australia, but we would expect it to be similar.

This manual is designed for bee breeders – those who produce queens either for commercial sale or their own business. The first section outlines the theory behind modern animal breeding. The second section covers how to implement modern animal breeding in your business, and provides data collection templates and best-practice advice. The third section contains the trait dictionary; the guides and photos enable you to score traits. Data recording sheets for use in the field and files to transfer this data electronically are available to download by clicking the hyperlinks where they are presented below. As modern bee breeding continues to develop in Australia, we anticipate this manual will be updated to reflect this; the manual should be considered a starting point to be developed further in consultation with industry, rather than the end point.

Photo: Kirra Hughes

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General principles of modern breeding

Summary

- Genetic merit is a measure of how much the performance of a hive is due to the genetics of the queen through her progeny, rather than the environment and other factors.
- An estimated breeding value (EBV) estimates the genetic merit of a queen, using all available information (including performance of relatives).
- For an EBV to be calculated, information on pedigree, performance and important environmental factors is required.
- A breeding objective drives the selection emphasis placed on different traits, and the direction of selection.
- Selection response is proportional to the selection intensity and accuracy, which predominantly depend on replacement rates, trait heritability, the amount of data collected, and the accuracy with which genetic merit is estimated. Annual response is also affected by generation intervals.

Introduction

In its most basic sense, breeding works by harnessing the genetic variation in a population to produce offspring with desired traits. Animals within a breed vary in performance and in underlying genetic make-up (sometimes referred to as genetic merit, or quality). Genetic merit can differ among bees for any performance trait or other characteristic, such as honey production, temperament, pest and disease incidence, or size.

But how do you measure a colony's performance?

While differences between hives are observable without exact measurement of traits, it is difficult to quantify the variation in performance between colonies without data. Variation in performance (phenotype) between colonies reflects the environment (including management) and their genetics (Figure 1). Different traits vary in how much they are affected by the environment versus genetics. Improvement through breeding can act only on the portion of variation that is due to genetics.

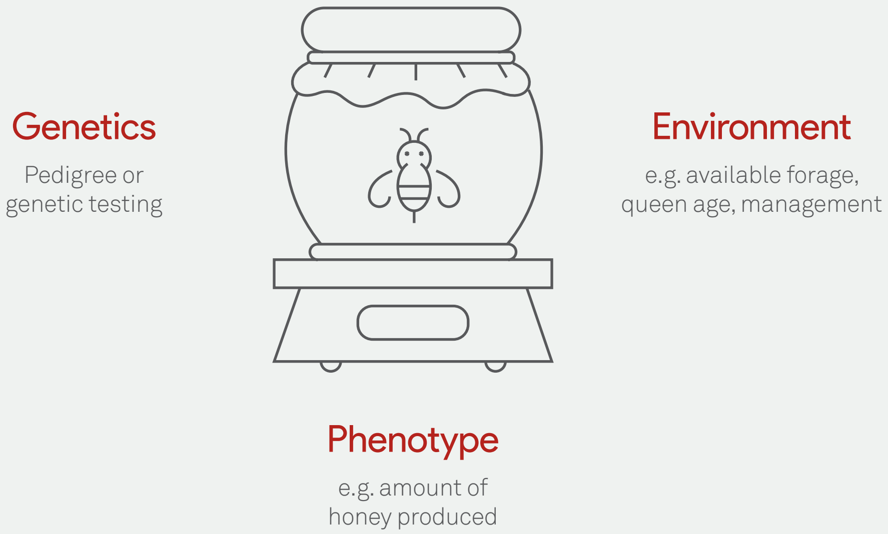


Figure 1. The phenotype of the hive. In this example, the amount of honey produced is influenced by the genetics and the environment (which also incorporates non-environmental factors, such as management differences).

Genetic merit

The phenotype of a colony is a blend of the genetic make-up of the colony and the environmental factors (which are not always identifiable) that influence each trait or phenotype. Genetic merit describes how much of the difference in the observed performance (phenotype) between colonies is due to the genes of the queen influencing her own characteristics, and those of her workers. This is the true value of the queen, and we can estimate genetic merit only from appropriate data.

The environment of a hive includes such things as the beekeeper's management, the load the hive belongs to, the apiary the hive is in, forage availability and quality as influenced by the location of the apiary, the weather, and the age of the queen (e.g. influencing population dynamics and egg and pheromone production).

We do not know the true genetic merit of an individual. Instead, we estimate its merit from collected data. An estimated breeding value (EBV) gives the numeric estimate of the performance of a queen for a specific trait compared to the population average for that trait. A positive EBV means the queen has high expression of the trait; a negative EBV means the queen has low expression of the trait. An EBV will represent only the genetic component contributing to the trait of interest; non-genetic effects are accounted for when calculating EBVs.

In the most sophisticated genetic evaluation systems, a queen's EBV does not come only from information collected on her colony, but also information about her relatives. If her relatives perform well, this will elevate her EBV; if they perform poorly, this will reduce her EBV. Relationships between animals can be established using pedigree and/or genotypic data. You don't need to, and won't be able to, calculate EBVs yourself (Banks *et al.* 2020). However, it is important you understand how they can be used. This manual is designed to help along every step of the breeding cycle (Figure 2).

The breeding cycle

Training

You should start by reading this manual, reviewing other Plan Bee material, and signing up for any training events. You can also contact the manual authors to talk through what you want to achieve.

Develop standard operating procedures (SOPs) for your workplace, specifying the equipment and methods to be used, and how to record and store data. Also, train all personnel to ensure they understand the correct procedures and can implement them consistently.

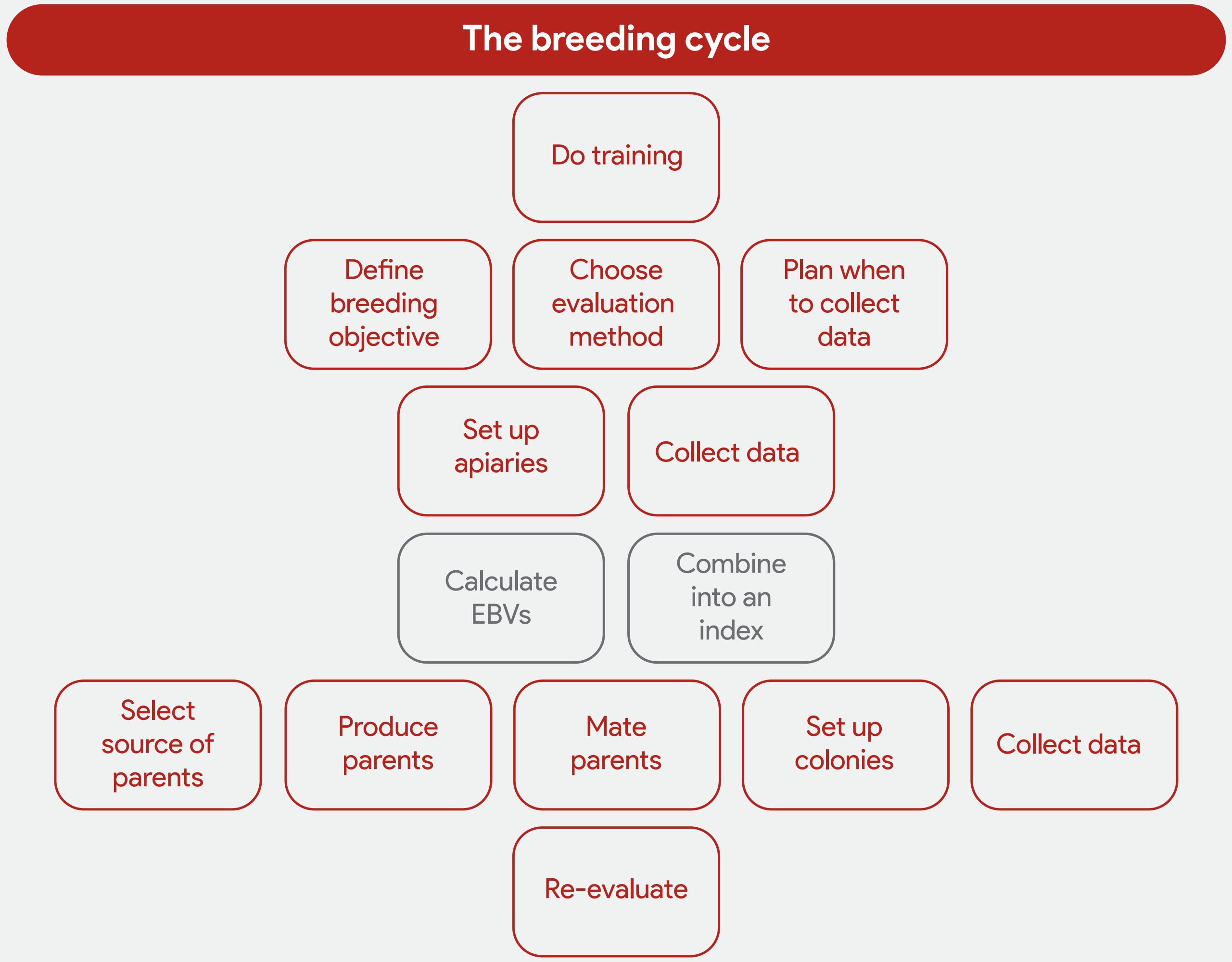


Figure 2. The breeding cycle – the pathway you will take to implement modern animal breeding into your business. Steps in grey will be performed by the Animal Genetics and Breeding Unit.

Define breeding objective

Your breeding objective is the overall aim of your breeding program. It includes the traits you will select for, the direction in which you want to select, and the relative importance (weight) given to the traits. Be aware that producing a 'super bee' with superior performance in every trait is impossible. A modest increase in specific desirable characteristics, however, is achievable through selective breeding. These improved characteristics must be continually selected for every year to maintain rates of improvements for desirable traits.

The traits you will select

When designing your breeding objective, you will first need to pinpoint the traits you want to change. The more traits under selection, the less progress that can be made on each one. Australian beekeepers are interested in many traits, but relatively few are of high priority (Chapman and Frost 2021, 2022, 2023). In your breeding objective, Plan Bee recommends including traits such as honey production, aggression, population, brood pattern, and pest and disease incidence.

All hives with American foulbrood must be euthanised according to the Bee Biosecurity Code of Practice (BBCoP) (Plant Health Australia 2022); this removes them from the breeding pool. Due to the varroa mite incursion, there are more requirements not covered in the BBCoP and these may be subject to change. Please refer to industry communications for relevant updated advice.

You might exclude queens as candidates for selection because of other traits. Decide on them. When a queen is excluded, it is important to record that decision so that the information is transferred to the Plan Bee database.

For example, if queens are excluded because their colonies have sacbrood, you should record that these colonies have sacbrood, and that other colonies do not. Without this recording, the information is lost and cannot contribute to EBVs for sacbrood, for example. This will help to identify the genetics that contributed to not having the disease in the rest of your population.

As another example, you might want to exclude any colonies from selection that have a chalkbrood score greater than 3 (i.e. 4 or 5) at any evaluation.

To be included in a breeding objective, traits should meet the following criteria (Figure 3):

1. Be economically important

This means they contribute to higher income and/or lower costs, or easier production.

2. Be heritable

There must be variation between colonies for the trait. Some of this variation must be genetic in origin so that offspring can inherit these differences.

Heritability is the proportion of variation in a trait that can be attributed to genetics, rather than the environment or other non-genetic factors affecting trait expression. Heritability can be defined simply as the proportion of observed difference in performance that is passed on to progeny.

For example, if all queens have exactly the same genetic make-up but some are bigger than others, the difference must be due to the environment. In this instance, the bigger queens might have been better fed – you cannot select for a bigger queen because there are no genes for being bigger. However, if there are genes for being bigger, then, given the same environment, some individuals will be bigger than others. That is, given the same environment, some individuals will perform better because of variations in their DNA (genotype).

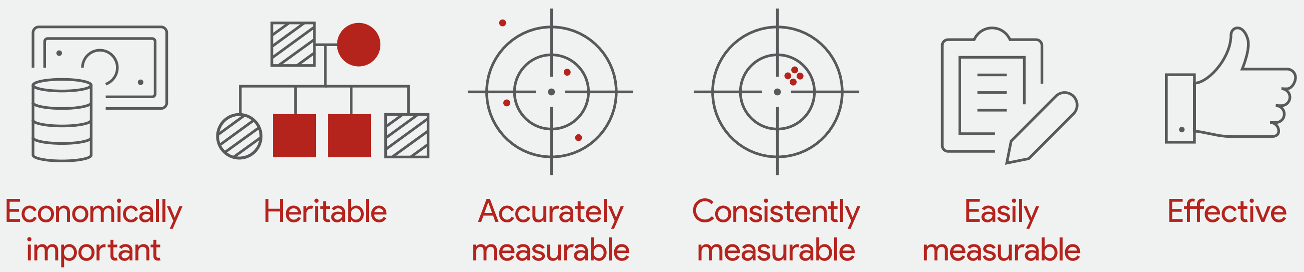


Figure 3. Six considerations when determining your breeding objective.

A heritability of 0.3 indicates that 30% of the phenotypic variation (after removal of known non-genetic factors) is due to variation in the genes that control the trait.

Traits vary in the relative importance of genetics (heritability) on their expression, which affects how easy it is to select for a trait. In Figure 4, if the three circles represented three different traits, the third trait would be more influenced by genetics than the other two; it would be easier to select for the third trait than the second trait, which would be easier to select for than the first trait. Heritability is not static; it changes with the environment and varies between populations. If the circles in Figure 4 represented the same trait in three different populations, there would be a stronger genetic effect in the third population, making it easier to select for that trait in that population, and more difficult in the first two populations. Heritability is also affected by how accurately you can measure the trait.

Heritability is estimated by measuring traits and determining the association between the values recorded for the trait (the phenotype) and the level of relatedness between individuals recorded for that trait (the genotype).

3. Be accurately measurable

Accuracy describes how well we can measure a trait. Are we close to the true value? For example, you could measure honey production as the number of supers of honey produced or the weight of the honey extracted from each colony. More accurate measurements (e.g. the exact amount of honey extracted) will result in more accurate comparisons and more accurate breeding decisions.

4. Be consistently measurable

Measurement consistency can also be referred to as repeatability. A measurement is repeatable if each time you evaluate a colony you get the same phenotype, and/or the ranking of different colonies is consistent across measurements. The top colony doesn't always have to rank as the top, but it should be consistently among the top performers. Different people assessing the same colony for the same trait at the same time should record consistent data. If the data you collect is completely random, you will not be able to make good breeding decisions using the data.

Some traits, including temperament and hygienic behaviour, are heavily influenced by the environmental conditions at the time of evaluation. Take care to evaluate such traits under uniform conditions. Repeat the assessment several times to account for the influence of low repeatability or the impact of environmental variation on the phenotypes.

5. Be easily measurable

Do you need specialist training? Do you need specialist equipment, such as a microscope or liquid nitrogen? Does measurement take a long time? Selection for genetic change is most effective when all colonies can be easily assessed, and therefore compared, for important traits. If a method to measure a trait is too difficult, bee breeders may choose not to evaluate that trait. There is often a trade-off between measurement difficulty and measurement accuracy. The quality of the data is important; just because something is easy doesn't mean it is worth investing your time in measuring it.

6. Have the desired effect

Some pairs of traits have common genes, which means that if you select for one of the pair, you will see correlated change in the other. For this to be useful, the traits must be reasonably strongly correlated (i.e. they share many genes in common). If they are not reasonably strongly correlated, selecting for a trait other than the one you are directly interested in will not achieve the desired result.

Figure 4. Representation of the genetic (red) and environmental (grey) effects for different traits in the same population, or the same trait in different populations.

An example is selecting for a short brood development period to provide resistance to *Varroa*. Having a shorter brood development period means less time for varroa mites to mature. It is possible to reduce the length of time taken for bees to mature. However, this does not mean there is less need to treat bees for varroa mite (Holmes *et al.* 2024). Selecting for shorter brood period is not effective at breeding *Varroa*-resistant bees because, genetically, brood period is not very strongly related to *Varroa* resistance.

The direction in which you want to select

Bee breeders can choose the direction in which they want to select traits. Typically, the direction of selection for a trait is obvious – either to increase or decrease manifestation of that trait. There are some traits, such as pest and disease incidence, that all beekeepers would agree to decrease. However, they disagree over the direction of selection for other traits, such as propolis and pollen production. This is why our scoring system is based on 1 (low) to 5 (high) – it avoids the use of judgement terms and enables everyone to be placed on the same scale. We need to know what the data is telling us, which is possible only if the data is linked to a quantitative, rather than a qualitative, measurement.

Alternatively, bee breeders can select for a particular score. If a trait is scored on a scale of 1 (low) to 5 (high), breeders might want, for example, a medium score (3). Depending on the current performance of a population, this might mean no selection is required, or alternatively, selection is up or down to achieve the intermediate performance. This also illustrates the importance of using a 1 (low) to 5 (high) scale. If beekeepers felt that an average amount was best, they would give these colonies a 5 if the scale was based on quality, and we wouldn't know whether a 4 meant there was too much or not enough. We then wouldn't be able to compare the data from this beekeeper with that from another who thought a low or high amount of this trait was best.

Relative importance of traits in a breeding objective

Typically, traits are weighted on their economic value and the genetic parameters that define how heritable they are and how they are associated with each other. In honey bees, we do not have good information on the economic value of many traits, and we also have few estimates of genetic parameters; honey is an exception. For example, how much is temperament worth to a business? Providing the relative importance (weight) of a trait tells Plan Bee the economic value of the trait to your business. It is possible for a trait to be very important economically, but if it is not heritable (i.e. partially controlled by genetics), it will not be part of your breeding objective.

Choose measurement method

There are several measurement options for most traits in this manual. While some are better than others, we recognise that not everyone will agree which is best for them. Our recommended method is in Section 3. Importantly, many colonies must be evaluated using the same method to produce EBVs. If you choose an unpopular method to evaluate your trait, you might not get EBVs for that trait because there is not enough data to perform the analysis.

Plan when to collect data

Determine the frequency, time and conditions under which you will evaluate colonies.

The more times each colony is recorded, the more data you will contribute to estimating differences between colonies, and the more accurate the EBVs will be. As ever, you must balance the time taken to record traits against the information doing so will generate. There may also be a trade-off between the number of colonies you assess and the number of times you assess them.

The BBCoP (Plant Health Australia 2022) requires that you assess colony strength and pest and disease presence on all hives at least twice a year; each assessment must be completed at least four months apart. This creates an opportunity to record data for these traits, but we welcome more data collections. There may also be regulations relating to how often you must check for varroa mites, which are subject to change.

The length of the beekeeping season varies across Australia, and the somewhat unpredictable floral resources makes it difficult to set industry-wide timings for evaluation. Evaluation timing also depends on when new queens are produced because colonies cannot be assessed for a minimum of six weeks afterwards. We suggest that you start by mapping out when colonies can be evaluated, then lock in times for a general/biosecurity evaluation (which always includes a *Varroa* infestation check) to meet your BBCoP requirements (Figure 5).

In Figure 5, the length of the beekeeping season varies for each beekeeper. Beekeepers (a), (b) and (e) set up new colonies with new queens in spring and must wait at least six weeks before performing their first evaluation. Beekeeper (c) does not requeen and so does not have a turnover period and can assess queens over the whole season. Beekeeper (d) requeens in autumn, so could move their evaluation earlier in the season if they wished. Honey will be extracted as the floral resources dictate, and the number of extractions will vary between beekeepers. More *Varroa* infestation checks may be legally required; they are also recommended to help you determine whether *Varroa* treatment/management is required, and to test efficacy.

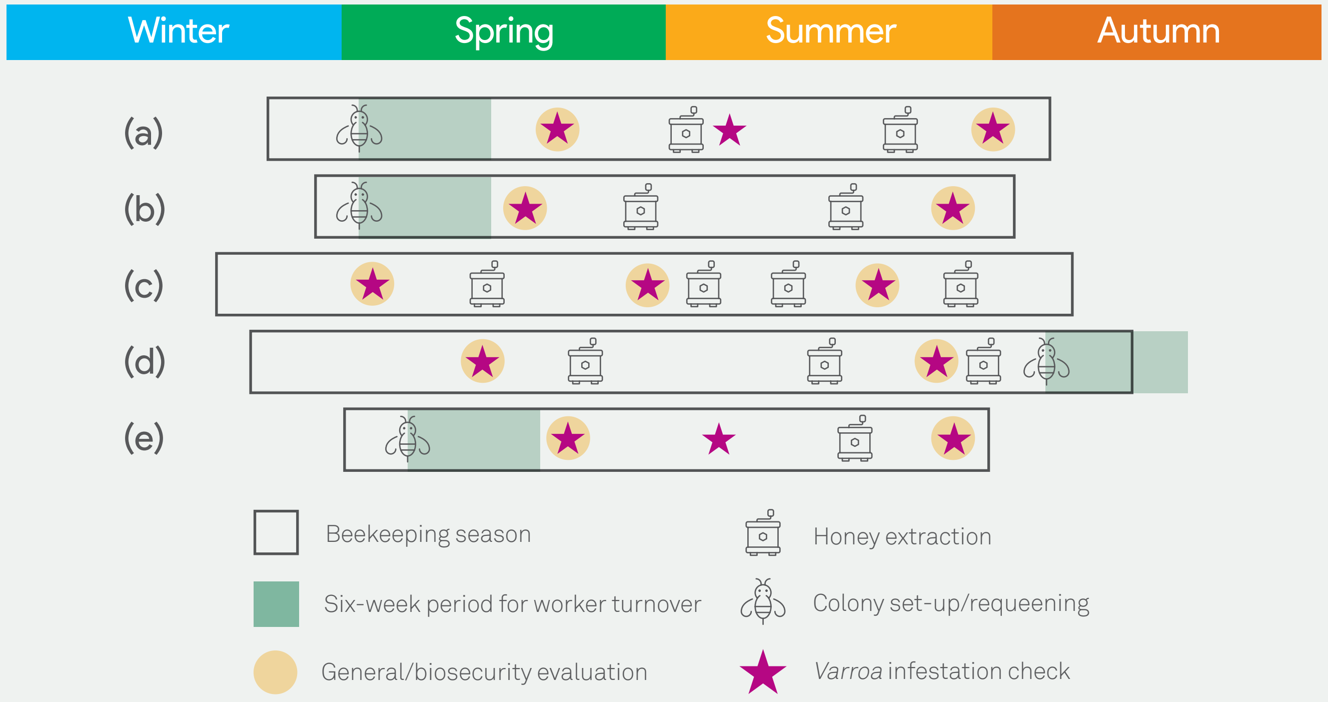


Figure 5. Examples of data collection timings for one season for different beekeepers (a-e). Please check current regulations relating to how frequently you must check for *Varroa* infestation.

Set up apiaries

Label your hives

It is imperative that data collected on a queen is tied to that queen. This means knowing which hive she is in, which is impossible unless you put individual labels on your hives. To do this, beekeepers are using cattle tags, QR codes, metal plates or engravements. Further, it is particularly important to connect the data to the right queen. A hive recorded before requeening reflects the resident queen, whereas a hive recorded after requeening reflects the new queen. Providing dates for new colony set-up and requeening and each recording event is crucial. While the hive label might not change, the colony (bees) within the hive changes when a new queen is introduced. The Plan Bee database will define colonies based on hive labels and information on the queen. Büchler *et al.* (2024) suggest that the best place to label a hive is on the bottom board because these are rarely changed. Alternatively, label the brood box – it is unlikely to be removed from the hive.

Mark your queens

It is important that you record when a queen is superseded. This means you need some means to differentiate your queen from any new queen that the colony may raise itself – you could clip her wings, paint a dot on her thorax, or apply a queen tag (coloured and numbered tag). If the resident queen has been superseded without human assistance (i.e. the hive has swarmed, or it killed or lost its queen and replaced her with a daughter), supply a meaningful date of this exchange and create a new, unique identity for the new queen.

Typically, artificially inseminated queens are labelled with queen tags. In this case, record the number on the tag on her thorax to link to the queen's unique ID and hive ID.

Spread genetics across apiaries and loads

To maximise the accuracy of comparisons among queens, it is good practice to ensure there is diversity of genetics across your apiaries and loads (Büchler *et al.* 2024). Do not keep all queens from one line together in a single apiary, and those of another line together in a different apiary, or all daughters of one queen in a single apiary if you have more than one apiary. This will make it difficult to separate genetic effects from environmental (in this case apiary) effects, given that each apiary provides an opportunity for large environmental differences to occur. Apiaries should consist of a minimum of 10 colonies (Büchler *et al.* 2024).

Introduce the queen

Data should not be used in the six weeks after a queen is replaced because this is when the hive transitions to reflect the new queen's genetics.

In the European breeding program (SmartBees), new queens are introduced to an artificial swarm (shook swarm) that has a standardised weight of worker bees (Uzunov *et al.* 2015). This gives all queens an equal start; thus, differences between colonies are more likely to be due to genetics than the environment. **This is best practice**. Use at least 2 kg of bees, which is standardised across the operation, and provide the hive with frames of foundation (Büchler *et al.* 2024).

An alternative to creating artificial swarms for requeening is to standardise the hives before introducing new queens. Ensure they have similar amounts of food stores, brood frames and bees, and are free of disease. This will not be possible if you use a barrier management system.

The third option is to complete an evaluation before requeening so that you know the quality of the hive the queen is being placed in. If one queen is introduced into a weak, disease-affected hive and another is introduced into a strong, healthy hive, it will appear that one queen is outperforming the other when it is possibly the other way around. In Figure 6, the colony in the orange hive was suffering from lots of chalkbrood and had a lower population and less stored honey than the colony in the blue hive. The beekeeper requeened both hives. After the queens had established (minimum six weeks), evaluation revealed that the blue hive colony had a lower chalkbrood score, more honey and an equal number of bees to that of the orange hive colony. The beekeeper might therefore consider the new blue hive queen to be the better queen. However, taking their starting point into consideration, the new queen in the orange hive has performed better – the colony produced 0.75 more frames of honey and two extra frames of bees than the blue hive colony over the period, while mostly cleaning up the chalkbrood.

Including the six weeks after requeening is problematic because part of the performance of the colony over this period is due to contributions made by workers of the previous queen. Therefore, after requeening, you might need to adjust the first record within the EBV analysis to account for the quality of the recipient hive at the time of requeening. This is possible only if that data is recorded. Simply replacing an old queen with a new queen may save you time and effort, but will reduce the benefit of the time you invest in evaluating colonies due to less accurate EBVs. We strongly recommend one of the other options.

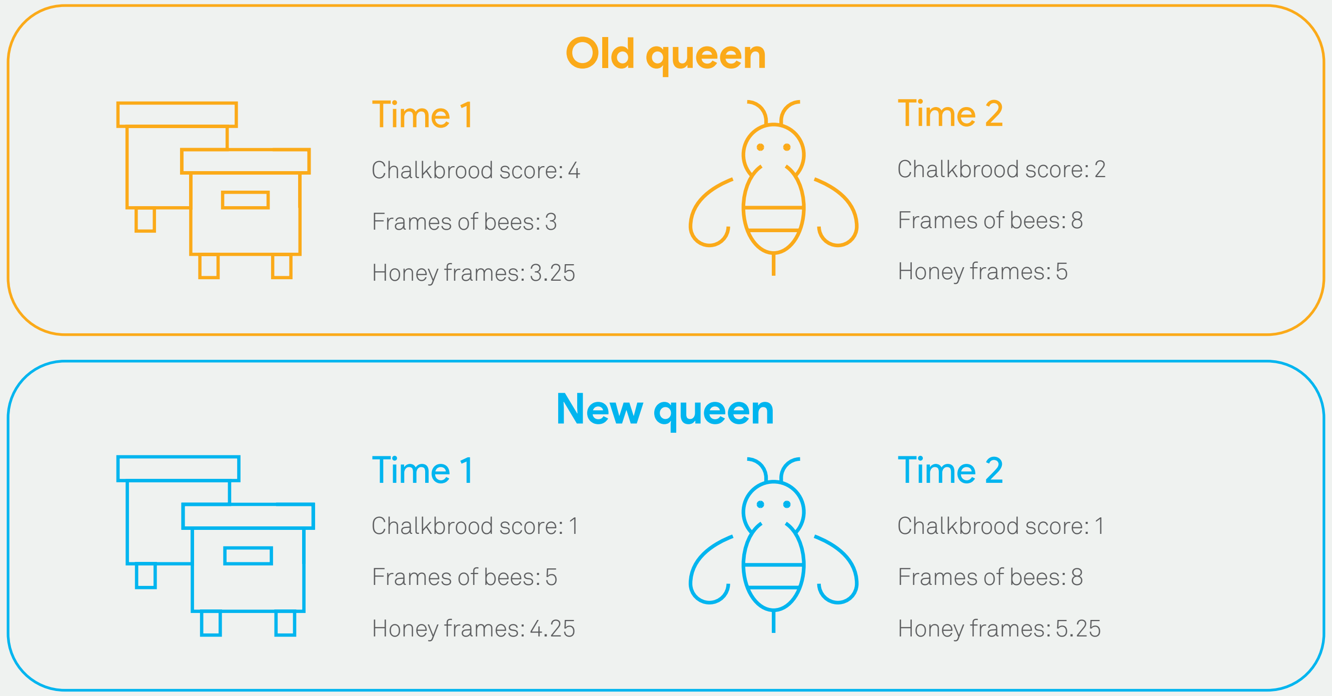


Figure 6. Example of colonies in two hives (orange and blue) assessed for chalkbrood score, frames of bees and frames of honey before and after having their queen replaced.

Collect data

Three types of data are required to produce EBVs: pedigree, performance and environment (Figure 7).

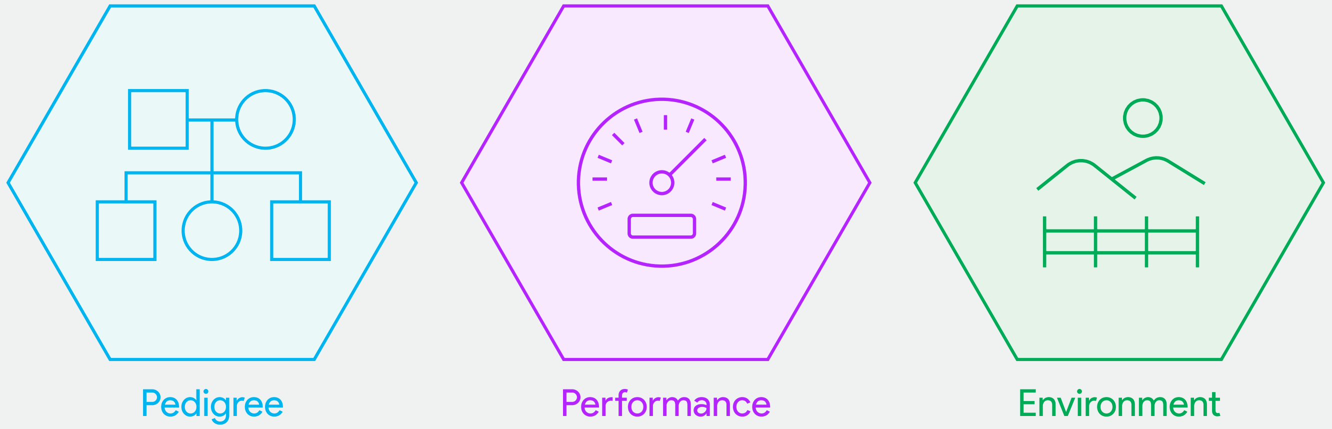


Figure 7. The three types of data required to produce an estimated breeding value.

Pedigree

Each queen in the database requires a unique queen ID and information on her pedigree. Pedigree represents the genetic relationship of one colony with another. To calculate EBVs, it is imperative to know the relationship between queens for which performance data is gathered.

Because a queen passes on genes to her daughters, the performance of her daughters will reflect these genes. A group of sister queens should perform more similarly than unrelated queens because they have common genes.

Most genetic evaluations are based on describing the relationships between individuals via a pedigree (progeny, queen mother). In bee breeding, the pedigree will commonly reflect the dam lines (queen mother and her daughters) if drones are unknown.

At a minimum, we need to know the identity of each queen's mother. This enables us to determine which queens are sisters.

Complementary to recording known matings, obtaining genotypic data from genetic testing can improve the ability to describe the genetic relationships between individuals because it enables more accurate modelling of the proportion of genes individuals have in common. Having genetic data can also help track the male line, so we know which queens are full sisters and which are half-sisters.

Currently, genotyping (genetic testing) is costly and slow for honey bees. A bespoke system for genotyping Australian honey bees is needed, but will take time to develop. It is recommended that producers continue to record queen pedigree even if samples are collected for genotyping. In the medium term, gathering genotypic data will provide the opportunity to understand the relationship between queens and colonies where the historic pedigree is unknown. Currently, waiting for the results of genetic testing (genotyping) before getting EBVs and making breeding decisions may reduce the number of generations that can be selected in a season, and slow genetic progress. Keeping records of pedigree is essential.

Performance data

Performance data involves evaluating colonies for common traits using standardised methods in a systematic way, and recording the results. Traits proposed for Australian bee breeders are outlined in Section 3.

Environment

The performance of a colony for the phenotype you are recording reflects the environmental factors and the genetic merit of the queen. Environmental and other effects include factors we can describe or know about, and unknown effects that might influence only one or some of the colonies evaluated at the same time. We can never know all non-genetic effects that influence individual performance, but we can typically account for some important factors and improve the accuracy of comparisons made between hives. Adjusting for environmental effects in genetic evaluation is difficult because of the challenge of quantifying the impact of factors such as weather, access to pollen and location. Consequently, to account for these effects, the EBV analysis fits at minimum a contemporary group effect. A contemporary group is a group of colonies that have been managed under the same conditions; in most cases, this will be a load and/or apiary. To separate genetic effects from environmental effects, we need information on how the colonies are managed. This information will come from several sources (Figure 8).

Which colonies?

When evaluating colonies, you should aim to record all colonies that are managed together in a contemporary group, e.g. a load/apiary. The genetic evaluation is based on comparing the performance of colonies that have been managed under the same environment. Consequently, selective recording of colonies has three main consequences: the analysis will unfairly classify the mean environment; the superiority of colonies may be underestimated or overestimated; and EBVs based on such information will be less reliable. Importantly, don't collect data only on your best colonies and no data on other colonies.

In Figure 9, a beekeeper has two apiaries, each with 24 hives. In the first, the colonies produce an average of 55 kg of honey, while those in the second apiary produce an average of 48 kg. If the beekeeper choses to collect data only on the top 10 colonies (those that are outlined) in each apiary, they would find an average of 63.9 kg for both apiaries, and wrongly conclude there is no effect of apiary. Now look at the top colony in each apiary – blue in the first and green in the second: they both produce 71 kg. Based on only the top 10 colonies, they both produced 7.1 kg above their apiary average and are considered equally as good. However, when taking information from all colonies into account, we find that the colony in the blue hive has produced 16 kg above the apiary average, and the colony in the green hive has produced 23 kg above the apiary average. There is a clear winner, but the wrong decision could be made if the full range of colony phenotypes was not recorded.

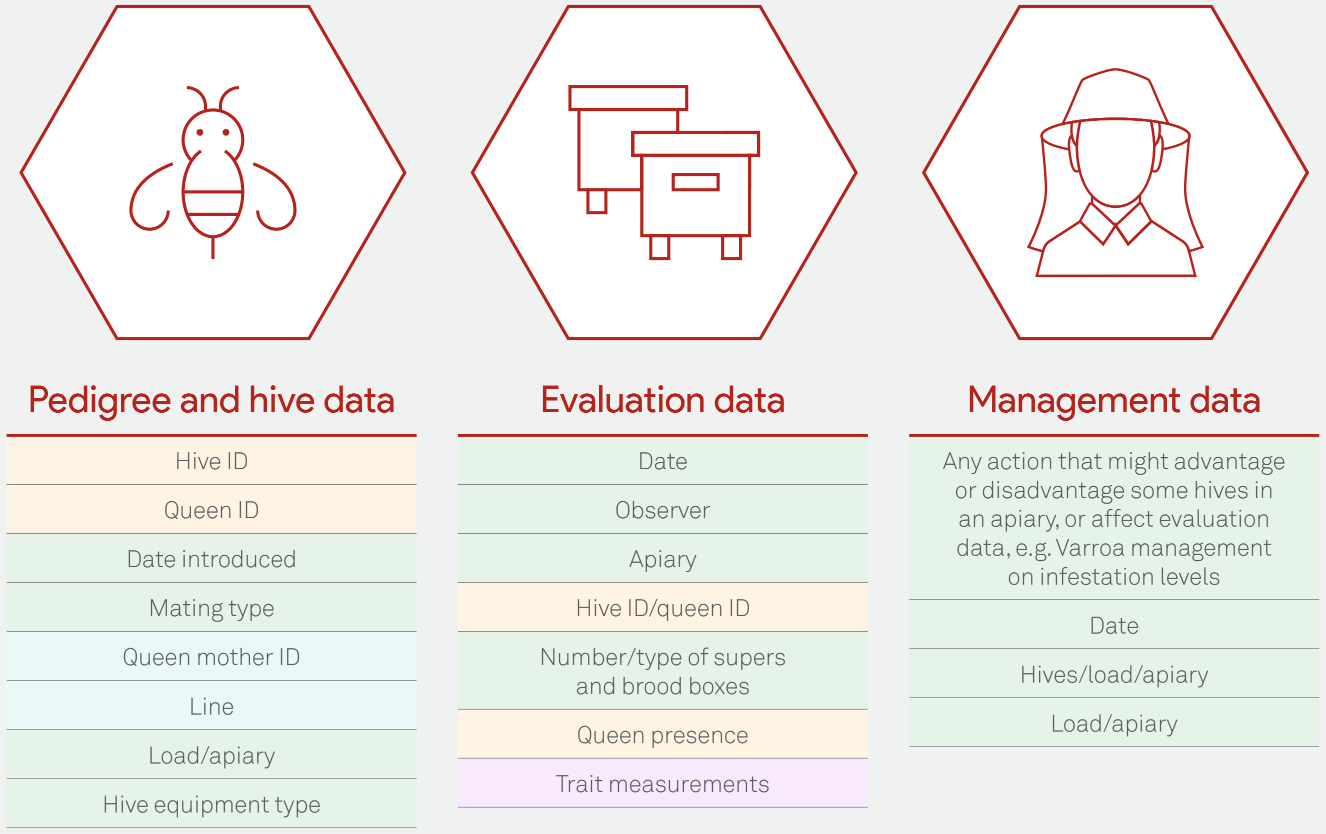
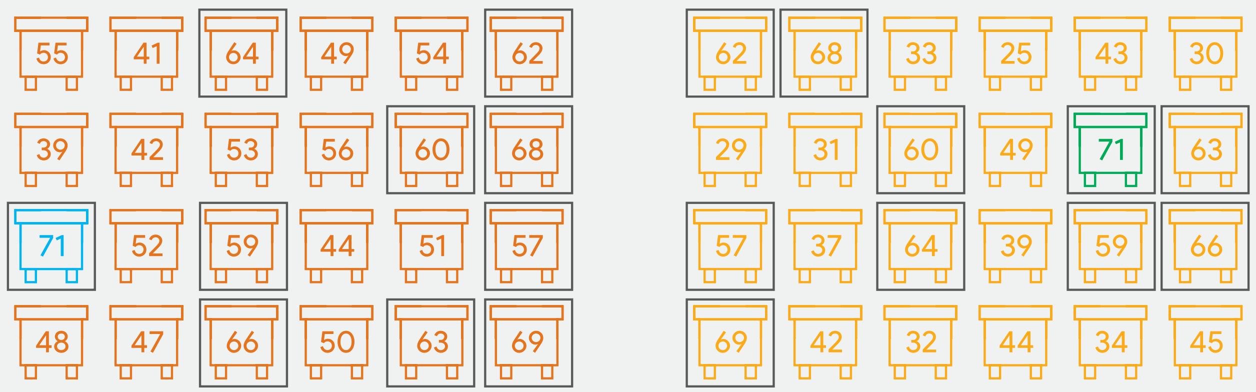


Figure 8. Three types of data are required to produce estimated breeding values: pedigree and hive data; evaluation data; and management data. Data recording sheets are provided in this manual to collect data on pedigree (blue labels), performance (purple labels), environment (green labels) and general information (yellow labels).



Apiary average: 55 kg
Top 10 colonies average: 63.9 kg
Blue colony based on apiary average: +16 kg
Blue colony based on top 10: +7.1 kg

Apiary average: 48 kg
Top 10 colonies average: 63.9 kg
Green colony based on apiary average: +23 kg
Green colony based on top 10: +7.1 kg

Figure 9. A beekeeper has two apiaries (orange and yellow), each with 24 hives. Honey (in kilograms) produced by each colony is noted on each hive. The top 10 colonies in each apiary based on honey production are outlined; the top colony in the orange apiary is highlighted in blue, and the top colony in the yellow apiary is highlighted in green.

Consistency/reproducibility

Traits that must be scored introduce another complication because scoring is subjective. Reproducibility refers to how similarly two people would score a trait. If one person's average score for a trait in an apiary is 4, but another person's average score for the same trait is 2, then the scoring has low reproducibility. It is difficult to know whether the colonies that the second person inspected were truly worse, or if they applied different criteria to their scores.

Suggestions for scoring reproducibility are:

- Have the same person evaluate the trait in all colonies in the apiary.
- Use the 1 to 5 scoring system provided, given that the level associated with each score is already standardised.
- Inspect colonies in a random order to randomise the effect of time of day or the way your actions with one colony affect the behaviour of nearby colonies.
- Inspect when feasible, remembering that more inspections will result in more accurate EBVs.
- Refer staff to your standard operating procedures to ensure they evaluate in the same way.
- Evaluate colonies in good weather – between 15 °C and 30 °C, no rain, low wind.

Submit data

Contact elizabeth.frost@dpi.nsw.gov.au for guidance.

Calculate estimated breeding values

The base step in obtaining EBVs is to measure the performance of every hive in a group of hives that are in the same location at the same time, and record their pedigree and/or take drone samples for genetic testing. As an absolute minimum, you must record at least 10 hives in the same location at the same time.

To produce EBVs across industry, ideally at least 1,000 hives will have been recorded, enabling the heritability of each recorded trait to be calculated. If only one person was to record, EBVs could be produced, but they would allow comparisons to be made only within that breeder's operation. More generally, after a suitable number of hives have been recorded and pedigreed (and genotyped), EBVs can be calculated. The standard calculation method is best linear unbiased prediction (BLUP). Where genetic data (genotype) is available from genetic testing, it is termed gBLUP or, more generally, genomic selection.

EBVs are calculated from:

- Pedigree and performance data
- Pedigree, genotype (from genetic testing) and performance data
- Genotype (from genetic testing) data without performance data.

Note, however, that to deliver EBVs for breeders who submit tissue samples only for genetic testing, the procedure depends on a sufficient number of records of pedigree, genotype and performance from other breeders and/or research and development populations.

Each of these methods will consider the genetic relatedness of the queens. This means that estimates of the genetic merit of the queen will reflect not only her colony's performance, but also the performance of other queens and colonies that are genetically related to her. Related individuals have common genes and thus, from a genetics perspective, should perform similarly. In addition, BLUP analyses are easily extended to more than one trait, such that genetically correlated traits can inform EBVs for both traits.

The core unit for calculating EBVs is a group of colonies that have been in the same location/s at the same time – the contemporary group. These shared environmental effects are accounted for by determining the average for each load/apiary combination and adjusting the trait values to account for environmental differences (which are a combination of effects from, for example, weather and food resources). Adjustments for other effects will also be considered. For example, a recently requeened hive may not have the honey production of a well-established hive present in the same load due to a break in the brood cycle.

BLUP-based methods account for the accuracy of the information. An EBV based on comparing many colonies is more accurate than that for an apiary with few colonies. Similarly, a queen with more relatives with data recorded will have more comparisons contributing to her EBV.

The EBV gives the genetic merit of the queen for that trait. An EBV of 3.78 kg means that her colony produced 3.78 kg more honey than the average because of the queen's genetics. She passes half of her DNA onto her daughters; they get the other half from their father. Not knowing the EBV of the father, we assume that the daughter will produce 1.89 kg (3.78 × 0.5, where the 0.5 indicates that the daughter receives half her DNA from her mother) more honey than the average due to the genes she received from her mother.

Combine individual EBVs into an index

Single-trait selection is generally not practised in modern breeding programs; typically, more than one trait is economically important. The selection process is simplified if information from all traits can be combined using a selection index, which summarises merit for all traits into one value, which is then used as the basis for selection.

A simple index can be constructed using the economic weighting for each EBV, where these are known or indicated via the breeding objective.

Development of a more complex index requires data on genetics parameters (heritability/correlations) and economic values for traits included in the breeding objective. Genetic correlations occur when an increase in one trait is associated with an increase or decrease in another trait due to the same gene/s or gene pathways being involved. An example is European foulbrood and chalkbrood incidence; a hive may have genes that make workers better able to detect brood diseases and remove infected brood. Phenotypic correlations are similar, but are the result of genetic and environmental effects. For example, large population is associated with greater honey production because of a larger foraging force. While genes may be involved in both traits, there is a simple logistical reason for an increase in one being associated with an increase in the other, but the genetic association may be low.

Until data and parameters become available, and the economic values of different traits are established, Plan Bee provides no customised index for selection. This work will be ongoing as we establish the traits that can be reliably recorded and evaluated for bee breeding programs.

Select parents

Based on the EBV and selection index reports, pedigree and/or genotypic data, select queens to produce the next generation – queen mothers and drone mothers. Your selection is likely to be guided by your mating strategy. However, the best design for a bee breeding program is relatively poorly established and requires research. The ideal breeding program design depends on the number of hives, the required replacement rates (and therefore selection intensities), the ability to control mating of selected parents, the breeding objective (i.e. how many traits, and their relationships with each other) and overall logistical constraints. Complex design issues are typically examined using simulation (computer modelling) and evaluated for response to selection (i.e. rates of genetic gain) and the accumulation of inbreeding.

Produce parents

Raise new queens from queen mothers under standardised conditions, ensuring the following: a good population of nurse bees; larva of the appropriate age (12-24 hours old) is grafted; sufficient nutrition; and hygienic equipment (Frost and Somerville 2016; Büchler *et al.* 2024). Transfer queen cells to mating nucs (small colonies for the purpose of mating queens) one to two days before emergence (Büchler *et al.* 2024).

Raise drones from drone mothers under standardised conditions, ensuring sufficient nutrition and provision of drone comb (Frost and Somerville 2016; Büchler *et al.* 2024). Drones develop from egg to sexual maturity in 40 days, so drone production should start at least two months before mating takes place (Büchler *et al.* 2024). A queen excluder at the entrance will prevent drones from drifting into the hive from other hives and will ensure pedigrees can be maintained. However, drone confinement may not be feasible in all Australian environments and careful attention to worker hostility to confined drones must be taken to prevent drone mortality. Be aware that drone fertility can be affected by exposure to agrichemicals and the presence of *Varroa destructor* and other pests and diseases (Büchler *et al.* 2024).

Mate parents

Where drone populations for mating are not controlled, it is possible that genetically inferior drones will reduce genetic progress made through queen selection. Uncontrolled mating can reduce the rate of genetic progress by 47-99% (Plate *et al.* 2019), meaning that no changes in performance due to genetics can be expected. Where drones are also sourced from genetically superior queens, this is less likely to occur. However, the risk of inbreeding will increase without some control of mating. Record as much information as you can about pedigree and mating strategy in the data recording sheet.

Artificial insemination

Artificial insemination enables total control of mating, such that the queen will not mate with unselected drones. It is therefore expected that genetic gain will be faster than for natural mating. However, this is a time- and expertise-intensive method that is used by only some bee breeders, and often only to produce breeder queens.

For artificial inseminations, the standard semen dose per queen is 8-10 microlitres, which represents eight or more drones (Gillard and Oldroyd 2020). Of course, this depends on the average semen load per drone and whether the collected semen is mixed (normally by centrifuge).

When artificial insemination is used to control mating and the drones are confined to their hive to guarantee their pedigree, record the queen that produced the drones used in the insemination.

If a queen is artificially inseminated with semen from a single drone, the pedigree of her offspring is straightforward. Similarly, if a queen is inseminated with pooled semen, the semen pool (e.g. semen pool G10) should be uniquely identified, and the source of drones that contributed to that pool should be recorded. When artificial insemination is used, a different combination of drone mothers could be chosen for each queen mother. This would avoid greater inbreeding.

Natural mating

Mating may also be controlled by providing drone mother colonies at mating sites, isolated by three things: (1) distance (on an island with no resident population of bees); (2) time, such as Horner's use of temperature and light-regulated rooms to alter mating flight times (Gillard and Oldroyd 2020); or (3) by having an entrance that can be used only by worker bees but is removed for queens and drones to fly at later times (Büchler *et al.* 2024).

The absence of honey bees at an isolated mating station can be verified by:

- Placing feed stations in the area before use and assessing whether any foragers feed at the stations (Büchler *et al.* 2024). If no foragers are seen after a minimum of three hours, you may assume there is a negligible number of beehives in the area. Feed stations should not use honey because it is illegal to offer honey to bees outside a hive in Australia (Plant Health Australia 2022).
- Placing test mating nucs without drones in the area to determine whether queens mate successfully (Büchler *et al.* 2024).
- Using drone ballooning (cigarette filters treated with queen pheromone attached to a hydrogen-filled weather balloon and manipulated with fishing line) in the afternoon in clear weather to verify the absence of drones (Scheiner *et al.* 2013; Büchler *et al.* 2024).

Finally, mating can be partly controlled by providing many drone mother colonies at non-isolated mating sites (drone flooding). This means that the queens will most likely mate with drones coming from the selected colonies, rather than surrounding colonies that belong to other beekeepers, or feral colonies in the area.

At a minimum, 12 drone mother colonies should be provided per 100 queens being mated (Frost and Somerville 2016), to a minimum of 20 for 500 queens (Büchler *et al.* 2024). Not having enough mates can be a key reason for queen failure. The number of mates has been associated with colony survival and honey production. Do not disturb mating nucs during the queen flight period (Büchler *et al.* 2024), which is typically between 1pm and 5pm in Australia. Ensure that mating nucs have adequate nutrition. All sister queens should be mated at the same mating station, so that they mate with the same pool of drones (Büchler *et al.* 2024).

Set up new colonies for evaluation

Once queens have mated (usually in mating nucs) and have been assessed for laying worker eggs and health (Büchler *et al.* 2024), give queens an equal start in life either by starting colonies as a shook swarm or equalising hives in terms of their food stores and bee population before adding the new queen.

Re-evaluate the program

Refine your recording practices. What worked and what didn't? How can you make things more efficient? Were you happy with the number of times you evaluated your colonies and the number of colonies you evaluated? Over 5-10 years, you may wish to re-evaluate your breeding objectives. Do you want to change your objective to investigate more or fewer traits?

Photo: Adobe Stock

Photo: Elizabeth Frost

2

Data recording

Queen ID

Every queen in the Plan Bee database requires a unique individual identification code. Using the hive ID in place of a queen ID is not satisfactory because queens are replaced within hives. Each queen must have a unique identifier in the database to ensure that data recorded only on that queen is considered when calculating her EBV. The Plan Bee queen ID is a combination of three codes: user code; age code; and generic code (Figure 10).

XXXX	YYMMDD	ZZZZZZZZZZZZ
User code	**Age code**	**Generic code**
Four numeric digits assigned to you.	Six numeric digits that detail the year, month and day, according to the format above.	1-20 alpha numeric digits.

Figure 10. The queen ID is made up of a user code, an age code and a generic code.

User code

This is a four-digit code starting from 1000. Plan Bee has assigned a user code to current participants.

Age code

This is a six-digit code: the year, month and day the queen was produced, in that order (YYMMDD). Where the day is not known, use 00. Similarly, if the month is not known, use 00.

For example, 241100 indicates the queen was produced in November 2024, but that the day is unknown.

Similarly, 230000 indicates the queen was produced in 2023, but the month and day are unknown.

The system is thus able to cater to different amounts of information. However, it is important to recognise that the age of the queen may be very important for some traits, so a minimum of year and month of production should be supplied, where possible.

There is likely to be significant uncertainty about the age of queens as queen breeders begin implementing the program. It is hoped that we will eventually move to knowing both the month and the year in which each queen is produced.

Generic code

This is what you use to identify the queen within your operation. The generic code can be up to 20 digits, combining letters and numbers. Note that it can be less than 20 digits.

You should refrain from using the name of the hive as the generic code. If for some reason you do not record the age code of the current queen in the hive, or the queen that you replace her with, then both queens will end up with the same queen ID (and colony ID, which is assigned in the database).

For example, your hive is labelled 1 and you use that as the generic code in your operation. Your user ID is 1000. You have no information on the queen's age. So the queen in the hive gets the queen ID 10000000001. At some time in the future, there is a different queen in the same hive box and again you have no information on the queen's age. So you end up with the same queen ID. Using the hive label as the generic code will work only if you always have enough information in your age code to differentiate queens when you have queen turnover.

Similarly, you might use a system, such as 23H17D, to show that the queen is the daughter of queen 23H17. This would work if two daughters were produced on different dates because the age code would differentiate between them.

However, this system would not work if multiple daughters were produced on the same day; additional notation would be needed to differentiate between the queens.

Each queen must have a unique generic code that never gets used again for another queen in your operation. This will ensure every queen in the Plan Bee database has a unique ID.

Previous queen ID

If you buy a queen from a queen breeder participating in Plan Bee, she should already have a unique queen ID that was assigned by the breeder, as described above. Use this previous queen ID when recording your own data so that she remains connected to her pedigree from the original breeder. **This is best practice.**

If you buy a queen from someone who has records on her in the Plan Bee database, but you cannot use her previous queen ID in your own data recording, you should record the previous queen ID in addition to your own queen ID. This will link data collected by you and the person from whom you bought the queen.

Pedigree and hive data

Essential information

Record the **queen ID**.

Record the **hive ID** of the hive the queen is in and update if/when she is moved. This hive ID must be unique within a beekeeping operation. There should not be two hives identified within a beekeeping operation as hive 101. As old queens are replaced with new queens, or new hives are built, they should be updated in the pedigree and hive information sheet. It is important that the same hive and queen can be returned to with certainty; this is possible only if the hives are labelled.

If you use loads, you must record the **load** the hive belongs to. A load is a group of hives that are always transported together and kept in the same apiary together.

You must record the date (YYMMDD) the queen was introduced to the hive, if it is known (**date introduced**). If you don't have all the information, give your best approximation or place zeros for the unknown information.

Record the **line** the queen belongs to. If you do not keep different lines, record the same code for all queens or leave blank. If you do keep lines but do not know the line of this queen, leave blank for that queen.

Record the **queen mother ID**.

Record information under **hive depth** and **hive width**, particularly if you have mixed hive dimensions. Use the same hive equipment for all hives (Büchler *et al.* 2024).

Use the following codes for depth:

- FD: Full depth
- H: Half-depth
- I: Ideal depth
- M: Manley depth
- 3Q: Three-quarter depth
- WSP: WSP depth

Use the following codes for box size (frames):

- 4: Four frames
- 5: Five frames
- 6: Six frames
- 8: Eight frames
- 9: Nine frames in a 10-frame box
- 10: 10 frames

Under **mating**, record whether the queen is artificially inseminated (AI), mated at an isolated mating station (I), or open mated (O).

If the queen was artificially inseminated, any known pedigree information on the drones and/or drone mothers should be recorded so that male pedigree can also be used. Typically, you will have information on individual drones only if they are collected for genetic testing. If you used pooled semen, and different pools for different queens, each pool should be given a label, and the drone mother IDs should be recorded against each pool.

A template has been provided (Table 1).

Optional information

If you use a queen tag, you may record it.

You may record the colour and race of the queen if you want to.

It is possible to record more detailed information. For the sake of clarity and to keep things simple, we have reduced the number of options covered in this manual compared to the first version.

Table 1. Pedigree and hive data recording sheet. Click here to download the Plan Bee data recording sheet (Microsoft Excel file).

Hive ID	Queen ID	Mating	Load	Apiary	Date introduced	Line	Queen mother ID	Depth	Width	Optional information/drone mother info/pooled semen ID
Example 1	2001231205E1H 2D1211R44	AI	Red	Home	240101	HL2	398222012422H2	FD	8	Pool 231211 – includes 20012202032 2HL4, 20012303 02HL3DR89, 20012212 00BL2C2321, 20012212 00BL5Y17 and 20012302 14SL3R24 Tag R44

Genetic testing

The purpose of genetic testing (also termed genotyping) for genetic evaluation is to obtain a queen genotype; it provides more detail on the genetic relationships among queens. A queen genotype can be established in two ways:

1. **Directly from the queen herself.** This method requires sacrificing the queen, which means that you cannot further evaluate her colony or continue to raise offspring from her. If she has already produced progeny, her genotype will inform the genetic evaluation system for her relatives (i.e. progeny) because it builds the reference between genotypes and phenotypes.
2. **Indirectly from her drones.** This can be done before her evaluation is complete to determine her suitability as a queen or drone mother. Estimating the queen's genotype in this way requires sampling 10 drone pupae. Using developing bees ensures they are the offspring of the resident queen (Figure 11). Drones are used because they come from unfertilised eggs; they carry only the queen's DNA. Having 10 of them increases our chance of getting both copies of the queen's alleles at each genetic marker to obtain her genotype.

Genotyping a small sample of worker bees will not accurately establish a queen's genotype because worker bees are themselves progeny of multiple drones.

Figure 11. Tweezers are used to pull samples from the drone cells and place them in a labelled tube. Photo: Nadine Chapman.

Sample collection

Keep samples as cold as possible to ensure the quality of the DNA. If the DNA degrades, the sample may not be usable. We will provide a kit that includes labelled tubes, tweezers and alcohol wipes.

Step-by-step instructions:

1. Uncap drone comb and assess the age of the available developing drones.
2. Choose the oldest developing drones, preferably with purple or darker eyes. We cannot guarantee DNA quality from any sample, but if samples are younger than this, it is unlikely that DNA quality will be sufficient for testing.
3. Using the tweezers, put all samples from a single hive in the tube. Use a different tube for each hive, and try to keep the head intact with the body.
4. Immerse the samples in the ethanol. The ethanol must penetrate the tissue to preserve it.
5. Record the details on your evaluation sheet – the tube ID against the hive ID or queen ID.
6. Wipe the tweezers with alcohol wipes or sanitiser wipes between using them for different hives. This ensures tissues from one hive do not contaminate samples from the next hive.
7. Keep the samples as cold as possible. If available, use dry ice. Alternatively, use ice blocks, ice cubes or a car freezer to keep the samples cold while working and during transit.
8. If the samples are on dry ice, keep them there because it is colder than your freezer. When the dry ice runs low, transfer them to your freezer. If you did not use dry ice, transfer the samples to the freezer.

Sample transport

Please don't post the samples. They will get hot and the DNA will degrade. We have several options for transport, depending on logistics:

- We will personally pick up the samples.
- We will organise for the samples to be couriered on dry ice.
- We will ask you to drop the samples off somewhere to be couriered on dry ice.

Sample processing

Samples are presently processed in batches according to the minimum number that will be processed by our service provider/s. The service provider processes the samples as they have availability. The genotypic data provided by the service provider is transferred to the Plan Bee database held by the Animal Genetics and Breeding Unit (AGBU), which will analyse it.

Trait evaluations

It is VERY important that all recorded data comes with either a queen ID or a hive ID, an apiary (location) and the date of recording, along with the staff and time where relevant.

Which hives?

When recording data, aim to evaluate all hives that are managed together in a contemporary group, e.g. a load/apiary. However, you may not be able to get through all hives in an apiary on the same day. We recommend that the contemporary group contains a minimum of 10 hives that are measured on the same day and by the same assessor.

The smallest unit we will consider is 10 hives. That means 10 hives in the same apiary, measured on the same day, by the same assessor.

Of course, we encourage you to evaluate many more than 10 hives because the accuracy of comparisons increases with the amount of data. If you have multiple assessors, it is best that one does the scoring while the other does the notetaking. This will improve consistency.

Partway through the season, you might make a shortlist of hives that are potential sources of replacement breeders. You might then choose to assess them for other traits that require a greater time investment. If needed, seek advice from someone at the AGBU. Firstly, it may not be possible to produce an EBV for the other traits because of insufficient colonies assessed for that trait. Secondly, a trait should be assessed on random subsets to capture the total variance within the population.

Some beekeepers have expressed a desire to use different methods for shortlisting and detailed evaluations – this should be avoided. Only data collected using the same method can be combined for analysis and production of EBVs.

Apiary records

We need to know which hives are kept together at the time data is collected. This enables us to separate genetic effects from environmental effects. You do not need to provide anything that identifies your apiary. For example, you can use an apiary code that is known only to you, as long as you use it consistently for that apiary. The BBCoP (Plant Health Australia 2022) requires you to record where your hives are located, so you should already have this information available.

Time

Track how long your evaluations take so you (and your customers) can better value it. Knowing how long it takes you to record data may help you to set fair prices for your queens. If time of day is important for a trait, then that can also be established from time-based data. An example is population – when the weather is cool, all bees will be at home; when it is warmer, many foragers will be out foraging.

Observer

The observer is the person who evaluates the colony. Recording the observer is essential if multiple people in your operation evaluate hives in the same apiary on the same day.

Hive ID or queen ID

You can record both the hive ID and the queen ID, but we need at least one of them on your data recording sheet. The two IDs will be matched together via your pedigree and hive information sheet, so it is important to keep these details up to date. Both IDs are needed for the database, but you do not need both on your sheet *if the information is recorded elsewhere and submitted together to the database.* If in doubt, record both IDs.

Hive make-up

The number of brood boxes and supers should be recorded, particularly if the data recorded can be affected by different hive configurations. This goes for different box sizes as well.

For example, the frames of bees present in a 10-frame hive will likely be higher than that in an eight-frame hive. If you run more than one hive configuration for hives you intend to compare, you must record this.

We strongly suggest you use a single box size in your operation (Büchler *et al.* 2024). If you use a mix of box sizes, you can either record this on your data sheet or in your pedigree and hive information sheet at the time of data recording.

Use the following codes for depth:

FD: Full depth

H: Half-depth

I: Ideal depth

M: Manley depth

3Q: Three-quarter depth

WSP: WSP depth

Use the following codes for box size (frames):

4: Four frames

5: Five frames

6: Six frames

8: Eight frames

9: Nine frames in a 10-frame box

10: 10 frames

For example, 3 × 10FD would mean three boxes of 10 full-depth frames.

Queen status

At the time of recording any traits, record the queen status. It is a secondary piece of information that identifies whether the correct queen is present at the time of recording, or whether this is uncertain. Use the following codes:

QR: Queenright, correct queen present and seen

NS: Not seen, but assumed present

QL: Queenless

V: Correct virgin present

C: Correct caged queen present

AINL: Artificially inseminated, not laying

SRV: Self-raised/supersedure, virgin

SRL: Self-raised/superseded, laying

DL: Drone layer

Don't forget to assign new queen IDs to new queens. If the colony has self-raised a supersedure queen, then the queen mother ID of the new queen is the old queen's ID. Please also record an approximate date the colony is likely to have superseded.

For example, a queen with ID 2001220927B6Y66P0310 gets superseded. Her daughter could take the ID 2001231000B6Y66P0310SR if you think that the supersedure date was sometime in October 2023, and you want the generic code to reflect that the queen is a self-raised daughter of the queen with generic code B6Y66P0310. Or the daughter could take another ID beginning with 2001231000, so long as no other queen has the same ID.

Tube

If you take samples for genetic testing, then record the number on the tube in your data sheet. You do not need to write the whole code. For example, if the code on the tube is 24PB356, you could write 24PB at the top of your sheet and then write 356 in the tube column. Please ensure we have all information we need to identify samples for testing – we need to be able to regenerate the entire tube ID.

Recommended traits

While your breeding objective is up to you, we recommend collecting data on the following traits as a minimum:

1. Frames of bees and brood pattern, representing population strength
2. Aggression, representing temperament
3. Chalkbrood, representing disease resistance
4. Honey production, representing production level
5. *Varroa* infestation, representing pest resistance (where *Varroa* is present).

Recording strategies

Creating routine recording strategies is important for bee breeders. Several types of beekeeping activities lend themselves to data recording:

1. Requeening or making new hives (record pedigree)
2. Recording traits during a BBCoP inspection (minimum of twice a year)
3. Working the hives under normal circumstances
4. Extracting honey
5. Checking for *Varroa* infestation.

General evaluation compliant with the BBCoP

The BBCoP (Plant Health Australia 2022) legally requires you to inspect hives twice a year, with inspections a minimum of four months apart. The Code also requires you to record hive strength and pest and disease incidence. Recording trait data during this industry-required activity is a way to make the best use of your time.

We have added the recommended traits to the general/biosecurity evaluation, which should take up to 10 minutes per hive.

Hive strength – required by the BBCoP; frames of bees to the nearest quarter-frame (e.g. 6.25, 6.5, 6.75, 7).

Pests and diseases – required by the BBCoP; either a score of 1 (low) to 5 (high), or the number seen.

Aggression – optional under the BBCoP but a recommended trait for a general inspection; a score of 1 (low) to 5 (high).

Brood pattern – optional under the BBCoP but a recommended trait for a general inspection; a score of 1 (low) to 5 (high), or the percentage of empty cells in a known area.

Table 2 is a template for a general/biosecurity evaluation.

Table 2. General/biosecurity evaluation recording sheet with the recommended traits and methods. Click here to download the Plan Bee data recording sheet (Microsoft Excel file).

Date		Apiary				Observer						Weather				
Hive ID	**Queen ID**	**Time**	**Super**	**Brood**	**Queen**	**Bees**	**BP**	**EFB**	**AFB**	**CB**	**Nos**	**Sac**	**SHB**	**WM**	**Vwash**	**Agg**
Example 2	2001231205B21H223	13:31	1×8 FD	1×8 FD	QR	6.75	4	1	1	2	1	1	1	1	1	1
Comments																
Comments																
Comments																
Comments																
Comments																
Comments																

Observer: Person doing the assessment; Weather: Overcast/sunny/windy; Hive ID: Hive ID; Queen ID: Queen ID; Time: Time you open the hive; Super: Number and type of supers; Brood: Number and type of brood boxes; Queen: Queen status; Bees: Frames of bees to one-quarter of a frame; BP: Brood pattern score; EFB: European foulbrood score; AFB: American foulbrood score; CB: Chalkbrood score; Nos: Nosema score; Sac: Sacbrood score; SHB: Small hive beetle score; WM: Waxmoth score; Vwash: Number of *Varroa* in an alcohol/soapy water wash or sugar shake; Agg: Aggression score. This inspection is compliant with the Honey Bee Biosecurity Code of Practice

Core trait evaluation

The recommended core traits for Plan Bee are honey production, hive strength, pests and diseases (chalkbrood, *Varroa*, small hive beetle), brood pattern, and aggression. You may want to include other traits; however, this will be beneficial only if there are enough records in the database for that trait. This evaluation is likely to take less than 10 minutes per hive. We have created a template for a core trait evaluation (Table 3). There is room for you to add traits if you want to. Honey production has not been included because it is evaluated when it is extracted.

Hive strength – frames of bees, to the nearest quarter-frame.

Pest and disease – chalkbrood (1-5 score), *Varroa* (number found in an alcohol or soapy water wash), small hive beetle (1-5 score).

Aggression – a score of 1 (low) to 5 (high).

Brood pattern – a score of 1 (low) to 5 (high), or the percentage of empty cells.

Honey extraction evaluation

Honey is typically harvested from hives more than once per season, and data from each harvest should be recorded as a new event. Each honey extraction event should be recorded separately. We have created a template for honey extraction evaluation (Table 4).

The time required varies on the set-up (for example, loader with weighing capacity, honey trolleyed onto scale platform before loading, or automated weighing of trackable supers in the honey shed before extracting).

If you inspect every hive per apiary when collecting honey, hives that had insufficient honey for harvesting on that date should receive a 0 kg honey record.

If the hive was NOT inspected to decide on honey harvesting, the hive should have no data supplied for honey harvest. Please record which hives were not inspected; without this information, we will not know whether the hive was not inspected or if it has been removed from the apiary and not recorded in your hive management recording sheet.

Hygienic behaviour or UBeeO evaluation

Hygienic behaviour testing using the freeze-killed brood (FKB) or pin-killed brood (PKB) method is a two-day procedure taking 10 minutes or more per hive on both days. Record results 24 hours after replacing the comb in the hive.

The UBeeO test (UBO) is a one-day procedure taking up to 10 minutes per hive. Record results two hours after replacing the comb in the hive.

While setting up the relevant recording protocol for hygienic behaviour, breeders have the opportunity to concurrently record other important core traits that may also be associated with the outcomes for hygienic behaviour. For example, record the strength of the hive and the nectar flow because it could influence hive performance. We have created a template for a hygienic behaviour/UBeeO evaluation (Table 5). Nominate the method you have used. Treat a minimum of 50 cells.

Hive strength – frames of bees.

Hygienic behaviour – FKB or PKB: percentage of removed cells. UBO: percentage of uncapped cells.

Mite non-reproduction evaluation

Inspect a minimum of 35 infested cells, but preferably 60. We have created a template for mite non-reproduction (Table 6). Evaluation of mite non-reproduction lends itself to assessing the infestation level of the brood, and the recapping level.

Hive strength – frames of bees.

Mite non-reproduction – percentage of infested worker cells with non-reproductive *Varroa*.

***Varroa* infestation** – percentage of worker cells infested with *Varroa*.

Recapping – percentage of infested cells recapped.

Varroa-sensitive hygiene evaluation

Inoculate a minimum of 35 cells with a single varroa mite and open but not inoculate the same number of cells. We have created a template for *Varroa*-sensitive hygiene (Table 7). Evaluation of *Varroa*-sensitive hygiene should include core traits that may affect this behaviour.

Hive strength – frames of bees.

***Varroa*-sensitive hygiene** – percentage of infested workers removed.

Varroa infestation check

The number of varroa mites is determined using one of three Plan Bee methods, some of which may not meet your surveillance requirements. A template has been provided (Table 8).

Table 3. Core evaluation data recording sheet. Click here to download the Plan Bee data recording sheet (Microsoft Excel file).

Date		Apiary				Observer					Weather			
Hive ID	**Queen ID**	**Time**	**Super**	**Brood**	**Queen**	**Bees**	**BP**	**CB**	**SHB**	**Vwash**	**Agg**			
Example 3	2001221000G45V12	10:41	2x9 FD	1x9 FD	NS	11	5	2	1	3	2			
Comments														
Comments														
Comments														
Comments														
Comments														
Comments														

Observer: Person doing the assessment; Weather: Overcast/sunny/windy; Hive ID: Hive ID; Queen ID: Queen ID; Time: Time the colony is opened; Super: Number and type of supers; Brood: Number and type of brood boxes; Queen: Queen status; Bees: Frames of bees to one-quarter of a frame; BP: Brood pattern score; CB: Chalkbrood score; SHB: Small hive beetle score; Vwash: Number of *Varroa* in an alcohol/soapy water wash or sugar shake; Agg: Aggression score. Blank columns left for additional traits of your choice taken from the trait dictionary.

Table 4. Honey extraction recording sheet. Click here to download the Plan Bee data recording sheet (Microsoft Excel file).

Date	Apiary	Observer	Weather

Hive ID	Queen ID	Super	Brood	Boxes	Frames	WeightP	WeightA	Hon	Comments
Example 4	2001231011Y12H16	1×8 I	1×8 FD	1		17.7	5.8	11.9	
Comments									
Comments									
Comments									
Comments									
Comments									
Comments									

Observer: Person doing the assessment; Weather: Overcast/sunny/windy; Hive ID: Hive ID; Queen ID: Queen ID; Observer: Person doing the assessment; Super: Number of supers; Brood: Number of brood boxes; Boxes: Number of supers of honey removed; Frames: Number of frames of honey removed where not part of a full removed honey super; WeightP: Weight pre-extraction; WeightA: Weight after extraction; Hon: Weight of honey extracted (WeightP – WeightA).

Table 5. Hygienic behaviour/UBeeO recording sheet. Click here to download the Plan Bee data recording sheet (Microsoft Excel file).

Test type (circle): **FKB** **PKB** **UBO**

Date	Apiary	Observer	Weather	Nectar flow

Hive ID	Queen ID	Super	Brood	Bees	Time 1	Cells 1	Empty 1	Time 2	2	Comments
Example 5	2001231100DB39023	1×8 H 1×8 FD	1×8 FD	9.25	10:26	53	2	12:26	22	
Comments										
Comments										
Comments										
Comments										
Comments										

Observer: Person doing the assessment; Weather: Overcast/sunny/windy; Nectar flow: Rate the nectar flow (low, medium, high); Hive ID: Hive ID; Queen ID: Queen ID; Super: Number and type of supers; Brood: Number and type of brood boxes; Bees: Frames of bees to one-quarter of a frame; Time 1: Time the test started; Cells 1: Number of cells treated; Empty 1: Number of treated cells that do not contain brood at the time of treatment (empty or uncapped for UBO); Time 2: Time you recorded the result; 2: Number of treated brood cells that are empty (FKB), uncapped (UBO) or sealed (PKB) at the end of the test.

Table 6. Mite non-reproduction, brood infestation and recapping data recording sheet. Click here to download the Plan Bee data recording sheet (Microsoft Excel file).

Date	Apiary	Observer	Weather

Hive ID	Queen ID	Time	Super	Brood	Bees	Open	Inf	NonRe	RecInf	RecUninf	Comments
Example 6	2001231100B22C8SR	8:43	1×9 FD	1x9 FD	7.5	251	60	31	35	19	

Observer: Person doing the assessment; Weather: Overcast/sunny/windy; Hive ID: Hive ID; Queen ID: Queen ID; Time: Time evaluation begins; Super: Number and type of supers; Brood: Number and type of brood boxes; Bees: Frames of bees to one-quarter of a frame; Open: Number of inspected (opened) worker cells of appropriate age; Inf: Number of inspected cells with *Varroa* present (infested); NonRe: Number of infested cells where there are not *Varroa* brood of appropriate age to be mature when the bee emerges; RecInf: Number of infested cells with recapped wax cappings; RecUninf: Number of uninfested cells with recapped wax cappings.

Table 7. *Varroa*-sensitive hygiene data recording sheet. Click here to download the Plan Bee data recording sheet (Microsoft Excel file).

Date 1	Date 2	Apiary	Observer	Weather

Hive ID	Queen ID	Time 1	Super	Brood	Bees	Treat	Cont	Time 2	TreatR	ContR	Comments
Example 7	2001231100B22C8SR	14:03	1x9 FD	1x9 FD	8.75	35	35	14:03	24	2	

Date 1: The day *Varroa* are placed in cells; Date 2: The day cells are checked for *Varroa*, eight days later; Observer: Person doing the assessment; Weather: Overcast/sunny/windy; Hive ID: Hive ID; Queen ID: Queen ID; Time 1: The time evaluation begins on day 1; Super: Number and type of supers; Brood: Number and type of brood boxes; Bees: Frames of bees to one-quarter of a frame; Treat: Number of worker cells opened and inoculated with *Varroa*; Cont: Number of worker cells opened but not inoculated with *Varroa* (control); Time 2: The time of evaluation begins on day 2; TreatR: Number of treated cells where the pupa has been removed; ContR: Number of control cells where the pupa has been removed or died.

Table 8. Data recording sheet for *Varroa* infestation checks via alcohol/soapy water wash or sugar shake, or bottom board.
Click here to download the Plan Bee data recording sheet (Microsoft Excel file).

Date	**Apiary**		**Weather**	
Observer	**Method (circle):**	**Wash**	**Mat**	

Hive ID	**Queen ID**	**Number of *Varroa***	**Comments**
Example 8	2001240127B5N3	2	

Photo: Kirra Hughes

Management

The way you manage hives, collectively or individually, can affect their performance. If hives are treated differently before recording events, it is important to record these treatments so that queens are fairly compared. Where hives are not managed the same within an apiary, this should be recorded.

Be aware that supplementary feeding some of your hives or feeding them different amounts, or swapping food stores or removing/adding brood, means that hives are not on a level playing field – colonies will appear to have performed better when this is the result of your intervention. Manage all hives in an apiary the same way; this means not swapping frames of brood or food, giving the same amount of feed to all hives, and providing the same pest treatment to all hives in that apiary.

Please keep a management log (Table 9) of *Varroa* management and honey supering so that we can add this information to our analysis to produce more accurate EBVs. You may want to record *Varroa* treatments, supplementary feeding, small hive beetle treatments or the addition of supers, for example. This data will be used, where appropriate, to inform EBVs.

The way you manage hives, collectively or individually, can affect their performance. If hives are treated differently before recording events, it is important to record these treatments so that queens are fairly compared. Where hives are not managed the same within an apiary, this should be recorded.

Photo: Kirra Hughes

Table 9. Management recording sheet. Click here to download the Plan Bee data recording sheet (Microsoft Excel file).

Apiary	Load	Hives treated	Hives not treated	Treatment or management	Date applied	Date complete	Comments
Home	Example 9	G01-40		Bayvarol	240202	240329	

All hives in an apiary should be treated the same. When hives are treated differently, this should be recorded.

Checklist

- [ ] Hives have unique IDs.
- [ ] Queens have unique IDs that include information on their age.
- [ ] Queens are marked so that you know whether they have superseded.
- [ ] Queen introduction dates are known.
- [ ] Genetics are spread between loads (i.e. not all daughters of a queen are kept in the same apiary/load when you have multiple apiaries/loads).
- [ ] Queens within an apiary/load are provided with equal opportunity – they are managed the same, and started from shook swarms or were provided with hives of the same/similar strength and food stores when introduced.
- [ ] Colonies are not evaluated for at least six weeks after being formed or being requeened.
- [ ] Pedigree and hive data (Figure 8) is recorded in the recording sheet (Table 1).
- [ ] Management data (Figure 8) is recorded in the recording sheet (Table 9).
- [ ] The breeding objective is defined.
- [ ] Hive evaluations are planned around seasonal activities.
- [ ] Traits are evaluated at least twice over the beekeeping season.
- [ ] Traits are evaluated consistently throughout the season using the same method and following the scoring guides provided.
- [ ] Hive evaluation information (Figure 8) is recorded in the recording sheets (Tables 2-8).
- [ ] All hives in an apiary have been evaluated on the same day by the same person. Alternatively, at least 10 hives have been evaluated in the same apiary on the same day by the same person.
- [ ] Evaluations have included more than just the best hives.
- [ ] As many hives as possible have been evaluated.
- [ ] EBVs or a breeding index (and knowledge of genetics, if available to avoid inbreeding) have been used to choose the best queens to produce the next generation.
- [ ] Queens have been raised and mated using best practice, ensuring mating control.

Figures 7 and 8 have been repeated here for your convenience.

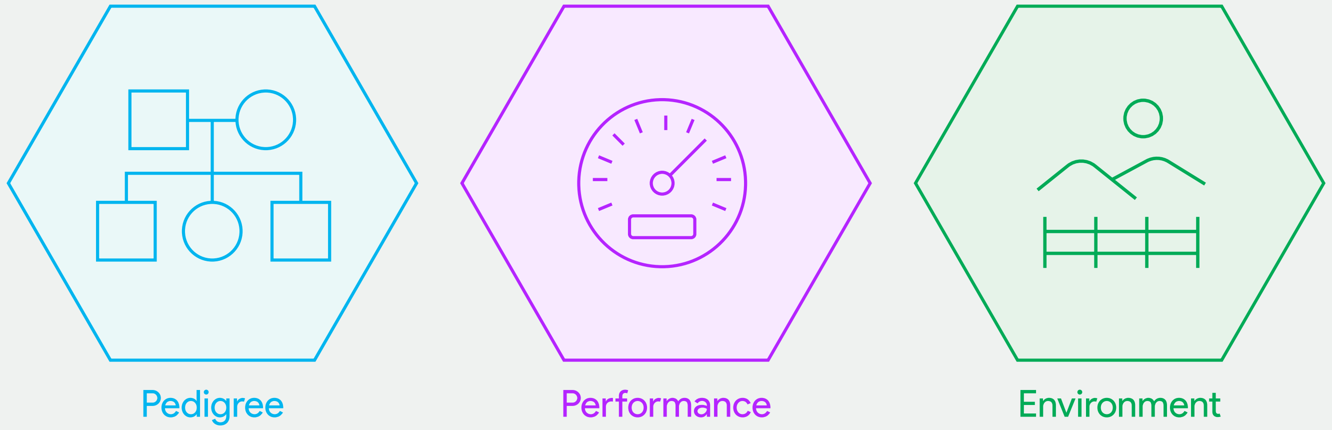
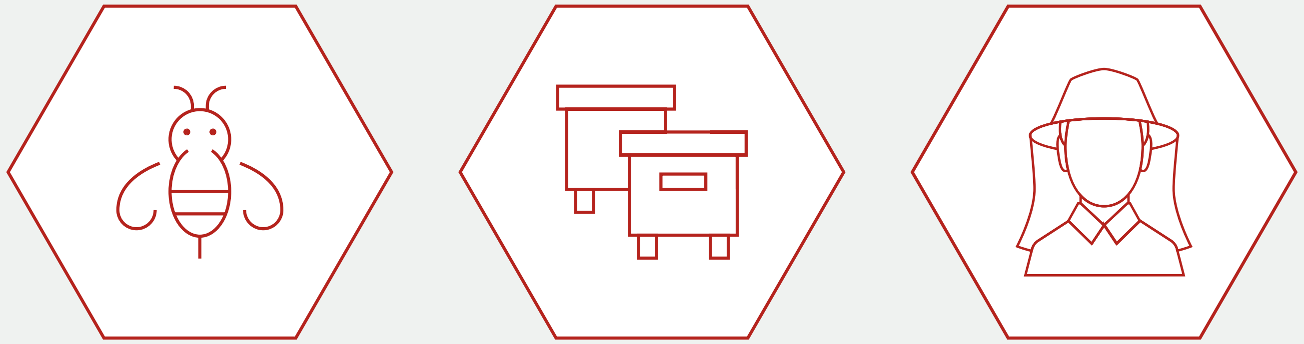


Figure 7. The three types of data required to produce an estimated breeding value.



Pedigree and hive data	Evaluation data	Management data
Hive ID	Date	Any action that might advantage or disadvantage some hives in an apiary, or affect evaluation data, e.g. Varroa management on infestation levels
Queen ID	Observer	Date
Date introduced	Apiary	Hives/load/apiary
Mating type	Hive ID/queen ID	Load/apiary
Queen mother ID	Number/type of supers and brood boxes	
Line	Queen presence	
Load/apiary	Trait measurements	
Hive equipment type		

Figure 8. Three types of data are required to produce estimated breeding values: pedigree and hive data; evaluation data; and management data. Data recording sheets are provided in this manual to collect data on pedigree (blue labels), performance (purple labels), environment (green labels) and general information (yellow labels).

Photo: Adobe Stock

3

Trait dictionary and trait evaluation methods

Introduction

A variety of methods have been provided for most traits to encompass the feedback we received from industry. Keep in mind, however, that a large amount of data is required for a recording method to produce an EBV. If you are using an unpopular method, you will need to collect data on many hives to generate useful information.

We have placed an asterisk (*) against the recommended traits. Where possible, we have provided descriptions for traits measured on a scale (Table 15), and photos are provided for some traits (Tables 16-18). Plan Bee recommends including honey production, aggression, brood pattern, pests (including *Varroa*), diseases and frames of bees in your hive evaluations.

Production

Numerous hive products (Table 10) are marketable (Somerville and Frost 2021).

Table 10. Trait dictionary of production traits and the options (a-c) for recording them.

Trait	Trait abbreviation A – preferred method	Trait abbreviation B	Trait abbreviation C
Honey	Weight of extracted honey: Hon*	Area to 0.05 (5%) frame: HonF	
Pollen	Area to 0.05 (5%) frame: PolF	1 (low) - 5 (high): PolS	Weight in pollen trap: PolW
Burr comb	1 (low) – 5 (high): Burr		
Propolis	1 (low) – 5 (high): PropS	Weight: PropW	
Royal jelly	Weight from 10 cups: RJC	Weight per hive: RJH	

Honey

Honey production can be influenced by management. For example, if a hive needs a super but is not provided with one, it will produce less honey than it otherwise would have. If all hives in an apiary suffer the same disadvantage, there will be no effect on the accuracy of EBVs. However, if only some hives are disadvantaged by not having new supers, they will appear to have lower genetic merit than they actually do. Thus, it is better to err on the side of caution and add a super.

Büchler *et al.* (2024) suggest that supers should be added when bees cover at least three-quarters of the combs in the brood box, and removed when bees occupy less than two-thirds of the combs in the lower super; however, these recommendations may not suit the Australian environment. If you do not have enough supers available for your needs, you should record that the hive needed a super in the management data recording sheet (Table 2) so that it can be considered during data analysis.

Younger queens tend to lay more eggs than older queens, resulting in a larger population and greater honey production (Hauser and Lensky 1994). Providing the queen age in your queen ID will ensure this variation is considered in analysis. Honey production can be measured by weighing, in kilograms, the amount of honey extracted from the hive (Hon*). All supers must have the hive ID recorded on them. Weigh the super before removing the frames. Mark all the frames from a super before extracting the honey. After extraction, return the same frames to the same super and weigh it. Alternatively, if you use the same box size on all hives, you may instead subtract the weight of an average box after extraction.

The weight (kg) of extracted honey is calculated as follows: weight of super(s) before extraction – weight of super(s) after extraction. The balance used should have an accuracy of 0.1 kg (100 g), be calibrated and be levelled (Büchler *et al.* 2024).

Stored honey/nectar can be scored to the nearest 5% of a frame in the field (HonF). This is done using a grid (Figure 12) (Büchler *et al.* 2024). Alternatively, you can mentally divide each side of the frame into 10 sections. In total, this gives 20 sections, so each section is 5% (0.05) of a frame. Estimate the area of stored honey to the nearest 5% for each frame and write the sum in your data recording sheet.

Pollen

Stored pollen can be scored to the nearest 5% of a frame (PolF) by using a grid (Figure 12) (Büchler *et al.* 2024). Alternatively, you can mentally divide each side of the frame into 10 sections. In total this gives 20 sections, so each section is 5% (0.05) of a frame. Estimate the area of stored pollen to the nearest 5% for each frame and write the sum in your data recording sheet.

Using the descriptions in Table 15 (PolS), score stored pollen from 1 (low/little) to 5 (high/heaps).

The amount of pollen collected in a pollen trap over 48 hours can be recorded in grams (PolW). Numerous pollen traps are available on the market (Somerville and Frost 2021). The same style of pollen trap should be applied to all hives but installing them could take considerable time. If so, empty each trap before starting your evaluation period and record the time that an empty trap was returned to the hive. Return 48 hours later. Place the collected pollen in a zip-lock bag labelled with the hive ID, using the same bag type for each hive. Weigh each bag and subtract the weight of an empty bag from the total of each hive; this will give you the amount of pollen collected in a 48-hour period. Ants can steal pollen, so you may need to control the ant population in the apiary (Somerville and Frost 2021).

Figure 12. A grid held over a frame to determine the area devoted to, for example, brood. Assess both sides of the frame to give the total area for that frame. Photo: Doug Somerville.

Burr comb

The amount of burr comb can be rated on a scale of 1 (low/little) to 5 (high/heaps) using the descriptions in Table 15 (Burr).

Propolis

The amount of propolis in the hive can be scored from 1 (low/little) to 5 (high/heaps) using the descriptions in Table 15 (PropS).

The amount of propolis can be recorded in grams (PropW). Place a propolis mat in the hive on top of the frames and under the lid. Place mats in all hives in the apiary on the same day. Mats will take some time to fill, depending on resources and the weather (Clarke 2019). Remove the mats on the same day, ensuring they are labelled with the hive ID. Place each mat in an individual bag to ensure no propolis falls off. Place the bags in the freezer (overnight or longer); doing so makes the propolis easier to dislodge from the mat. Collect the propolis into a container and weigh it, and record the weight minus the weight of the container.

Royal jelly

The weight of royal jelly collected from 10 queen cups/cells can be recorded in milligrams (RJC). Graft day-old female larvae into artificial queen cups/cells, then introduce the frame into the hive. Three days later, remove the frame, uncap the queen cells, and use a pipette, spatula or small spoon to collect the royal jelly, discarding the developing queen. There are also machines that can automatically harvest royal jelly (Clarke and McDonald 2017). Repeat for a further nine cups/cells, recording the total.

The weight of royal jelly collected from a hive can be recorded in grams (RJH). For the above method, however, the total amount collected from each hive is recorded. It is important that the same number of cups/cells is grafted per hive. Differences in production will occur due to the amount per cup/cell and the proportion of grafts that the hive accepts.

Temperament

Temperament (aggression and runny; Table 11) can be affected by several factors: environmental conditions; the weather; the time of day; the behaviour of nearby hives; and the way people handle hives (Büchler *et al.* 2024).

Record the time of recording and/or work hives in a different order each time you score for temperament to determine whether hives are naturally defensive or whether they've been triggered before being worked by being disturbed by other hives nearby.

Recording temperament multiple times across a season will make evaluations more accurate. It is also best if the same person or people record temperament for all hives being inspected that day to maintain consistency. If this is not practical, work a few hives together to ensure you would score them similarly when working separately on a future day. Büchler *et al.* (2024) suggest temperament traits should be evaluated on at least four separate occasions.

Table 11. Trait dictionary of temperament and appearance traits, and the options for recording them.

Trait	Trait abbreviation
Aggression	1 (low) – 5 (high): Agg*
Runny	1 (low) – 5 (high): Run
Yellow colour	1 (low) – 5 (high): Y
Black colour	1 (low) – 5 (high): B
Red colour	1 (low) – 5 (high): R
Grey colour	1 (low) – 5 (high): G
Worker length	1 (low) – 5 (high): WS
Drone length	1 (low) – 5 (high): DS
Queen length	1 (low) – 5 (high): QS
Queen weight	Weight: QW

Aggression

Aggression (amount of smoke needed, attacking) can be rated on a scale of 1 (low/gentle) to 5 (high/aggressive) using the descriptions in Table 15 (Agg*).

Runny

Runny (movement and clustering of bees) can be rated on a scale of 1 (low) to 5 (high) using the descriptions in Table 15 (Run).

Appearance

Appearance covers the size and colour of the bees (Table 11). Colour is often thought to indicate a hive's subspecies; however, the vast majority of honey bees in Australia are hybrids (Chapman *et al.* 2016). Based on the assumption that colour equals subspecies, beekeepers often assume the traits their bees have based on their colour. However, numerous genes are involved in colour, and these genes are unlikely to be closely linked with genes controlling one, let alone all, the traits that beekeepers are interested in.

If you keep bees of a particular colour, scoring colour can be useful as a quality control for how well you have controlled mating. If the queen's worker offspring are a mix of colours, but you provided only drone mother hives of the colour that you use, you know that the queen has likely mated with drones from outside your breeding program. Unless these drones were part of someone else's breeding program, it is likely they will be inferior to the drones that you supplied for the traits that you select for. As a result, the hive is likely to perform below expectations.

Yellow colour

Rated on a scale of 1 (fewer workers are yellow) to 5 (more workers are yellow) using the descriptions in Table 15 (Y).

Black colour

Rated on a scale of 1 (fewer workers are black) to 5 (more workers are black) using the descriptions in Table 15 (B).

Red colour

Rated on a scale of 1 (fewer workers are red) to 5 (more workers are red) using the descriptions in Table 15 (R).

Grey colour

Rated on a scale of 1 (fewer workers are grey) to 5 (more workers are grey) using the descriptions in Table 15 (G).

Size

Size varies between different subspecies, and is influenced by the available resources and the age of the frames the bees are reared in. If you want to select for size, it is best to provide new frames to control for the effect of the size of the comb on the size (length) of workers and drones.

The size (length and weight) of queens is largely determined by the age at which larva are grafted and the available resources. The size of the queen is important because larger queens have been found to store more sperm, have larger ovaries, and lay more eggs. Queen size is largely due to the age at which she was grafted, but there are genetic effects.

Worker length

Rated on a scale of 1 (low/little) to 5 (high/huge) using the descriptions in Table 15 (WS).

Drone length

Rated on a scale of 1 (low/little) to 5 (high/huge) using the descriptions in Table 15 (DS).

Queen length

Rated on a scale of 1 (low/little) to 5 (high/huge) using the descriptions in Table 15 (QS).

Queen weight

Queen weight can be recorded in milligrams (QW). Queen weight is most practical in an artificial insemination program at seven days after queen emergence. Measure queen weight when she has received her first dose of carbon dioxide and is immobile (Frost *et al.* 2021). Use an electronic balance with an accuracy of 0.1 mg (Büchler *et al.* 2024).

Brood nest

A number of traits can be assessed in the brood nest (Table 12).

Table 12. Trait dictionary of brood nest and health traits, and the options (a-c) for recording them.

Trait	Trait abbreviation A – preferred method	Trait abbreviation B	Trait abbreviation C
Brood area	Area to 5% frame: BroodF*	1 (low) – 5 (high): Brood	Area eggs to 5% frame: Fec
Brood pattern/viability	1 (low) – 5 (high): BP*	% empty cells: BV	
Average pollen area in brood frames	1 (low) – 5 (high): BNP		
Average honey area in brood frames	1 (low) – 5 (high): BNH		
Swarming	1 (low) – 5 (high): SwaS		
Larval feeding of workers	1 (low) – 5 (high): LF		
European foulbrood	1 (low) – 5 (high): EFB	# of infected cells: EFBN	
American foulbrood	1 (absent) – 2 (present): AFB		
Nosema	1 (absent) – 2 (present): Nos		
Chalkbrood	1 (low) – 5 (high): CB	# of infected cells: CBN	
Sacbrood	1 (low) – 5 (high): Sac		
Small hive beetle	1 (low) – 5 (high): SHB		
Wax moth	1 (absent) – 2 (present): WM		
Hive cleanliness	1 (low) – 5 (high): Clean		

Brood area

Brood area can be scored to the nearest 5% (0.05) of a frame (BroodF) by using a grid (Figure 12) (Büchler *et al.* 2024). Alternatively, you can mentally divide each side of the frame into 10 sections. In total, this gives 20 sections, so each section is 5% (0.05) of a frame. Estimate the area of brood (eggs, larvae and pupae) to the nearest 5% for each frame and write the sum in your data recording sheet.

Brood can be rated on a scale of 1 (low/little) to 5 (high/heaps) using the descriptions in Table 15 (Brood).

Fecundity can be measured as the number of eggs laid in a 24-hour period (Fec). Use a cage to confine a queen to a single empty frame for 24 hours. Use a grid (Figure 12) to estimate the area of eggs to the nearest 5% of a frame laid over this time. Alternatively, you can mentally divide each side of the frame into 10 sections. In total, this gives 20 sections, so each section is 5% (0.05) of a frame.

Brood pattern/viability

Brood pattern can be rated on a scale of 1 (low) to 5 (high) using the descriptions in Table 15 and the photos in Table 16 (BP*).

Brood viability can be measured by the number of empty cells in an area of 100 capped brood (BV).

Average pollen area in brood frames

The average area of pollen in the brood frames can be scored from 1 (low/little) to 5 (high/heaps) using the descriptions in Table 15 and the photos in Table 17 (BNP).

Average honey area in brood frames

The average area of honey in the brood frames can be scored from 1 (low/little) to 5 (high/heaps) using the descriptions in Table 15 and the photos in Table 18 (BNH).

Swarming

Swarm tendency can be scored on a scale of 1 (low/little) to 5 (high) using the definitions in Table 15 (SwaS). The highest score recorded for a hive in the evaluation season will be used for analysis (Büchler *et al.* 2024).

Larval feeding of workers

The amount of food fed to workers can be scored from 1 (low/little) to 5 (high/heaps) using the descriptions in Table 15.

Health

Multiple pests and diseases afflict honey bees (Table 12). It is possible to reduce pest and disease pressure by selecting queens whose hives keep pest and disease incidence low.

Disease

Estimate the number of infected cells on three brood frames. Incidence of disease – European foulbrood (EFB), American foulbrood (AFB), chalkbrood (CB*), sacbrood (Sac) and Nosema (Nos) – can be scored on a scale of 1 (low/little) to 5 (high/heaps) using the guide in Table 15. Hives with AFB must be destroyed.

The incidence of chalkbrood (CBN) and European foulbrood (EFBN) can be recorded as the number of infected cells on three brood frames. Remove a brood frame from the brood nest, shake or brush off the bees, and examine the cells for infection. Examine three brood frames in this way and record the total number of infected cells.

Pests

Incidence of small hive beetle (SHB*) and wax moth (WM) can be scored on a scale of 1 (low/little) to 5 (high/heaps) using the guide in Table 15. For *Varroa*, see the next section.

Cleanliness

Hive cleanliness (debris on bottom) can be scored on a scale of 1 (low/unclean) to 5 (high/clean) using the description in Table 15 (Clean). Lift the brood box off the base and examine the amount of debris.

Population

The population of the hive is associated with honey production and can be an indicator of health. Population at certain times of the year could indicate a hive's potential (Table 13). Varroa mites are highly attracted to drone brood, so if you are breeding for *Varroa* resistance, our analysis may be improved by having data on the amount of drone brood in a hive.

Table 13. Trait dictionary of population traits and the options (a-b) for recording them.

Trait	Trait abbreviation A – preferred method	Trait abbreviation B
Population	Area to one-quarter of a frame: Bees*	
Drone brood	Area to 5% of a frame: Drone	
Overwintering~	Ratio of frames of bees in early spring versus late autumn: WinBees	
Spring build-up~	Ratio of frames of bees in spring versus early spring: SprBees	Ratio of frames of brood to frames of bees in spring: SprR

~ These are calculated from information on other traits if performed at the applicable time. Therefore, they are not recorded; the traits used in the equations are recorded instead.

Frames of bees

Frames of bees can be scored to the nearest one-quarter of a frame (Bees*). Before using too much smoke, quickly count how many frames are fully covered with bees by looking down through the top bars. For a double (hive made of two boxes), bees on frames can be evaluated by looking down at the top bars of the frames of the bottom brood box and up through the bottom bars of the frames in the top box (Figure 13). A frame where bees fully cover both sides is recognised as one frame of bees; a frame can be measured to the nearest quarter (0.25, 0.50, 0.75) (Büchler *et al.* 2024). Ideally, count the population early in the morning before too many foragers have left the hive.

Drone brood

Drone brood can be scored to the nearest 5% of a frame (Drone) by using a grid (Figure 12). Alternatively, you can mentally divide each side of the frame into 10 sections. In total, this gives 20 sections, so each section is 5% (0.05) of a frame. Estimate the area of drone brood to the nearest 5% for each frame and write the sum in your data recording sheet.

Overwintering and spring build-up

The traits for spring build-up and overwintering (Table 13) are marketable traits that have been successfully selected for overseas (Büchler *et al.* 2024). Overseas measurements have been adjusted for Australian conditions. Over time, we will be able to determine whether these measures can offer early insight into how hives will perform over the season. Time your evaluations such that information on the amount of brood and number of bees can be used for selecting for spring build-up or overwintering ability, if you want to select for these traits.

Overwintering can be measured as the ratio of the adult population in early spring (when beekeeping season begins) versus late autumn (when beekeeping season ends) (WinBees). The calculation is: (frames of bees in early spring) ÷ (frames of bees in late autumn). Spring build-up can be measured as the ratio of the adult population in spring versus early spring (SprBees). The calculation is: (frames of bees in spring) ÷ (frames of bees in early spring). Spring build-up can also be measured as the ratio of the brood population in spring versus the adult population in spring (SprR). The calculation is: (frames of brood in spring) ÷ (frames of bees in spring).

Figure 13. Score 'frames of bees' by counting how many frames are fully covered by bees when you first open the colony and with minimal smoke. Photo: Erica Mo.

Varroa resistance

The focus of breeding for *Varroa* resistance should be to reduce mite numbers; a hive is more likely to survive if it has fewer mites (Holmes *et al.* 2024). A number of traits for *Varroa* resistance have been included in this manual (Table 14). Implementing *Varroa* resistance traits in the honey bee population is in its infancy in Australia, so more resources may be needed to train bee breeders in this approach. Optimally, a single method will be used nationally (and internationally) for each trait (Mondet *et al.* 2020a), and Australian beekeepers will collaborate to settle on the method that best suits our environment and style of beekeeping.

Australian regulations cover the thresholds of varroa mite infestation at which you must either manage (i.e. cultural or biological management strategies) or treat hives; please check current regulations because they are subject to change. All hives in an apiary should be treated/managed the same if you are trying to select for *Varroa* resistance, so that we can get an accurate picture of how each hive is dealing with *Varroa*. A current directive is that *Varroa* treatment/management actions must be recorded, but this might not always be the case. Sharing this data will enable Plan Bee to determine whether *Varroa* infestation data can be included to evaluate *Varroa* resistance.

Whichever trait(s) you choose to select for, we highly recommend that you determine *Varroa* infestation via an alcohol/soapy water wash or sugar shake at regular intervals. You can avoid repetitive strain injury by using automatic shakers. Over time, *Varroa* infestation data will enable you and Plan Bee to determine whether selection efforts are reducing *Varroa* numbers. This is essential for determining whether the stock is displaying resistance.

Table 14. Trait dictionary of *Varroa* resistance traits and the options (a-c) for recording them.

Trait	Trait abbreviation A – preferred method	Trait abbreviation B	Trait abbreviation C
Varroa infestation	As measured by alcohol/ soapy water wash or sugar shake: Vwash	As measured in worker brood cells: Vbrood	As measured on a sticky board in 48 hours: Vboard
Mite population growth	As measured by alcohol/soapy water wash or sugar shake: MPGwash	As measured from worker brood: MPGbrood	As measured on a sticky board: MPGboard
Varroa-sensitive hygiene	As measured from at least 35 treated and 35 control cells: VSH		
Recapping	As measured in 35 infested cells: RECinf	As measured in 200 cells: RECall	
Mite non-reproduction	As measured in a minimum of 35 infested cells: MNR		
Varroa grooming	As measured by damage to mites collected from the bottom board: Groom		
Hygienic behaviour	As measured with UBeeO: UBO	As measured with liquid nitrogen freeze-killed brood: FKB	As measured with pin-killed brood: PKB

Varroa infestation

Varroa infestation can be evaluated by the number of varroa mites detected in an alcohol/soapy water wash or sugar shake (assuming 300 bees/125 mL cup) (Vwash*).

Secondly, *Varroa* infestation can be evaluated by calculating the percentage of infested worker cells. The calculation is: (number of worker cells infested with *Varroa*) ÷ (number of worker cells inspected) × 100. This method is suggested for use in combination with a mite non-reproduction (MNR) or *Varroa*-sensitive hygiene (VSH) test. As of the time of writing, this is not an approved method for *Varroa* surveillance; if you use this method for breeding, you will need to use other methods to meet your surveillance requirements. We suggest you check the current requirements (Vbrood).

Thirdly, *Varroa* infestation can be evaluated by tallying the number of varroa mites trapped on a sticky board over 48 hours in combination with acaricide application (Vboard).

Mite population growth

Measuring mite population growth (MPG) involves measuring the infestation level at two time points. Harbo and Hoopingarner (1997) used a standard interval of 10 weeks, whereas it was monthly for the Russian bee breeding program in the United States (Rinderer and Coy 2020), and in Europe, every three to four weeks is suggested (Büchler *et al.* 2024). A wide variety of intervals was used in the different studies reviewed by Mondet *et al.* (2020a).

Varroa infestation checks are currently required at 16-week intervals in Australia, but that is subject to change. The BBCoP requires two inspections at least 16 weeks apart (Plant Health Australia 2022). To improve selection for this trait, it will be helpful if Australian bee breeders can agree on a standardised period between *Varroa* infestation checks to determine MPG; currently, 16 weeks seems to fit best with other requirements. Fact sheets and video guides for performing alcohol/soapy water washes, sugar shakes and sticky mat examinations are available from Plant Health Australia, your state government department of agriculture and www.varroa.org.au. Within your business, standardise the interval between time 1 and time 2. If industry adopts a standard interval, it should be used.

MPG is calculated as: (mites detected at time 2) – (mites detected at time 1). Mite populations can be evaluated using:

- Alcohol/soapy water wash or sugar shake of bees to fill a 125 mL cup (MPGwash).
- The number of worker cells in combination with a mite non-reproduction test (MPGbrood).
- A sticky board over 48 hours in combination with acaricide application (MPGboard).

Varroa-sensitive hygiene

Varroa-sensitive hygiene refers to the detection and removal of *Varroa* from brood cells (Mondet *et al.* 2020a; Holmes *et al.* 2024).

In the morning, take a frame containing many brood cells that are partially capped or close to being capped (Büchler *et al.* 2024). With thumbtacks, attach a transparent sheet to the frame. Mark the brood cells suitable for manipulation on the transparent sheet. To ensure the sheet is marked with the hive ID, remove the sheet and place the thumbtacks back in the frame to help reposition the sheet later. Return the comb to the hive.

In the afternoon, collect mites from heavily infested colonies using a sugar shake. Keep the mites in petri dishes with moist filter paper and use them as soon as possible (Büchler *et al.* 2024).

Six hours after cell marking, attach the appropriate transparent sheet back on the appropriate frame. Choose 70 cells that have been capped. Make a small hole in the capping and use a fine wet paint brush to place a single mite in the cell and push the cap back down gently (Büchler *et al.* 2024). Treat 35 cells in this way; another 35 cells should be opened and closed without introducing a mite to determine whether the technique is contributing to brood removal. Mark the cells infested with *Varroa* and the cells opened but not infested with *Varroa* with different colours on the transparent sheet (Büchler *et al.* 2024).

Eight days later, return to collect the frame. Place the transparent sheet back on the frame and record whether marked cells are capped, opened or emptied for control and treated cells. You can also investigate the caps to evaluate recapping (RECv) and the contents of the cells for MNR if you want. Photos of the steps involved in this procedure are available in Büchler *et al.* (2024).

Varroa-sensitive hygiene (VSH) can be calculated as:

$$\left(\frac{\textit{Number of removed infested cells}}{\textit{Number of treated cells}} - \frac{\textit{Number of removed control cells}}{\textit{Number of control cells}}\right) x\ 100$$

Varroa cell uncapping/recapping

Collect a frame with purple-eyed or older worker brood (i.e. minimum seven days post-capping). Use a needle, forceps or scalpel to cut around the edges of the cell cap, and gently detach the cap (Büchler *et al.* 2017). Examine the pupae and the cell for *Varroa*. Examine the underside of the cap. Recapped cells have a matte area where there is no silk (Figure 14).

A minimum of 35 infested cells of the correct age should be examined to determine the percentage of recapping of *Varroa*-infested cells (RECinf). The calculation is: (number of *Varroa*-inspected cells recapped) ÷ (number of *Varroa*-infested cells investigated) × 100.

Evaluate 200 cells per hive to determine the percentage of recapping of all cells (RECall). The calculation is: (number of recapped cells) ÷ (number of investigated cells) × 100.

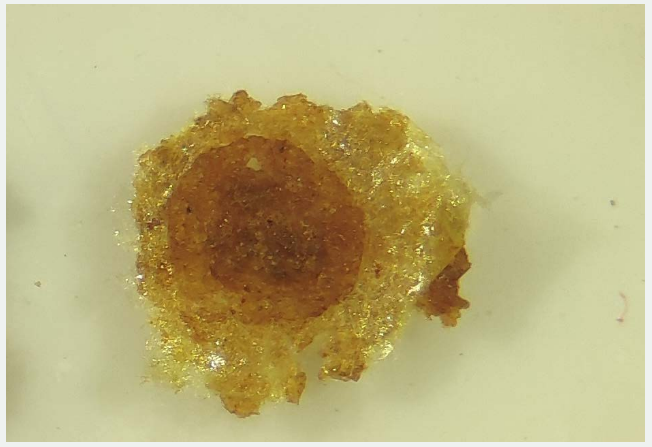


Figure 14. A worker brood wax cap that has been uncapped and recapped. The area that has been recapped is darker and not covered in silk. Photo: Michelle Liang.

Mite non-reproduction

Mite non-reproduction (MNR) may also be referred to as suppressed mite reproduction (SMR), and is sometimes referred to as *Varroa*-sensitive hygiene (VSH). VSH is one trait that contributes to the MNR phenotype (Holmes *et al.* 2024).

Test colonies should have no brood interruption or queen exchange for at least 60 days before testing (Büchler *et al.* 2024). Remove a frame of worker brood with purple eyes and white body, or that are older, from the hive. Gently open cells and assess the developmental stage of the worker brood as either less than seven days post-capping (eyes lighter than dark purple), seven to nine days post-capping (dark purple eyes and light body), or 10 to 12 days post-capping (dark eyes and dark body) (Büchler *et al.* 2017; Mondet *et al.* 2020b). Discard any that are less than seven days post-capping (Büchler *et al.* 2024).

Remove the pupae from the cell with forceps by sliding the forceps between the head and thorax and gently lifting up (Dietemann *et al.* 2013). Placed the pupae on a surface close to the cell to avoid dropping mites off the pupa. Use a light to inspect the cell walls for mites, and use a fine-tipped paintbrush to remove the contents of the cell. Assess the pupa and cell contents under a stereomicroscope (Dietemann *et al.* 2013). Record the stage of the eldest female offspring only when there is one foundress mite (cells have been infested by a single foundress) (Büchler *et al.* 2017; Mondet *et al.* 2020b). Daughters observed at 12 days post-capping may have the same appearance as a foundress (Büchler *et al.* 2017).

A foundress is considered reproductive if, when examined, one or more female offspring are at a stage of maturity such that she could be expected to be fully mature when the bee emerges. A foundress is considered non-reproductive if: (1) the female is too young to be mature when the bee emerges, indicating that the foundress's reproduction was delayed; (2) there was no male, therefore female offspring will be virgins; or (3) no offspring are present, indicating the foundress is infertile (Büchler *et al.* 2017; Mondet *et al.* 2020b). Young males can be difficult to distinguish from female protonymphs; males have longer, thinner legs and females have a rounder body shape (Büchler *et al.* 2017). Photographs of male and female mites of different development stages, and different stages of development of worker brood, are available in Büchler *et al.* (2017, 2024). Note, however, that Arista may use a different definition of reproductive.

Evaluate at least 35 (Büchler *et al.* 2024) but, preferably, 60 infested worker cells (Eynard *et al.* 2020). It may not be realistic to evaluate this trait at low infestation levels; if only 1% of brood cells are infested, it would require on average opening 6,000 cells to find 60 infested cells. It is unlikely there will be 6,000 appropriately aged (purple-eyed or older) pupae in the hive, and it would take a very long time to evaluate a single hive. When more than 30% of brood cells are infested with *Varroa*, you can expect to find 60 infested cells by opening fewer than 200 appropriately aged brood cells, or 35 infested cells by opening fewer than 120 appropriately aged brood cells (Figure 15).

Testing for MNR may be most suitable at peak *Varroa* population, which occurs after the hive has reached its peak bee population (Honey Bee Health Coalition 2022). Alternatively, bee breeders have other options: forgo treating and managing *Varroa* to get the population high enough to perform the test in a reasonable timeframe (subject to current legal requirements); deliberately infest hives with *Varroa* by collecting them using a sugar shake (subject to legal requirements); or allow frames of brood to be naturally infested by placing them in a host colony with high *Varroa* infestation (subject to legal requirements). If infesting hives, place the same number of *Varroa* in each hive in the apiary. Mites may take several days to enter brood cells, and brood need to be at least seven days post-capping. Thus, we suggest evaluating at least seven days after inoculating hives with *Varroa*. You may keep the caps to evaluate recapping (REC) if you want.

Where only pupae at least seven days post-capping are considered, and the standard definition of reproduction is used, MNR is calculated as: (number of non-reproductive single mite-infested cells) ÷ (number of single mite-infested cells) × 100. Where the Arista definition of MNR is used, the formula still applies.

Varroa grooming

Collect mites from the bottom board and assess them for damage under a microscope. Record damage to the legs or body, but not depression on the top of the body. Photos of damage are available from Hunt *et al.* (2016).

Grooming (Groom) is calculated as: (number of mites with damage) ÷ (number of mites collected) × 100.

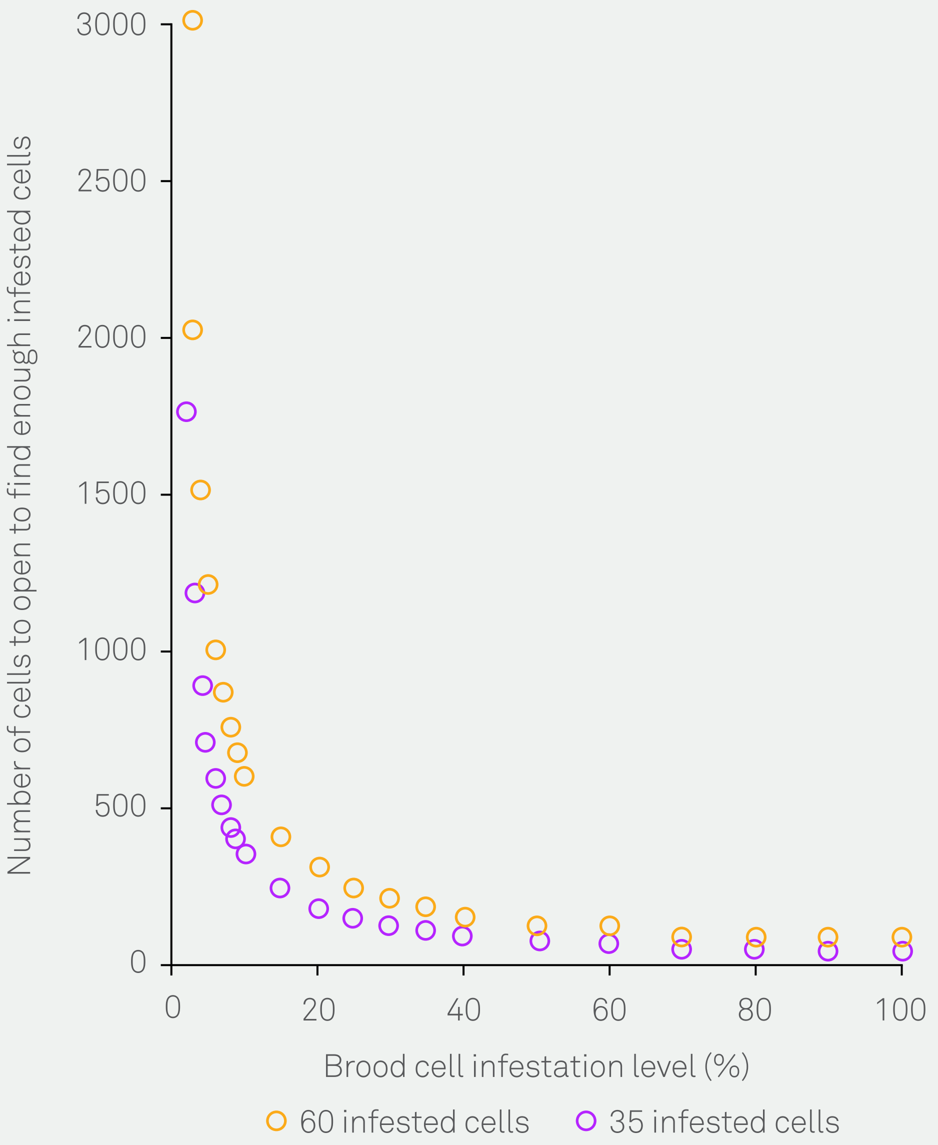


Figure 15. The number of cells that have to be opened to find either 35 or 60 Varroa-infested cells at different brood cell infestation levels.

Hygienic behaviour

There are three methods for testing for hygienic behaviour. Hives with a high response to UBeeO (UBO) are able to keep mite infestation levels low, which has increased survival (Wagoner *et al.* 2021). Selection for hygienic behaviour using freeze-killed brood (FKB) or pin-killed brood (PKB) is unlikely to reduce *Varroa* infestation levels below the point where you can forego treatment or management (Holmes *et al.* 2024). Selection for hygienic behaviour is also used to select for resistance to brood disease (Holmes *et al.* 2024). FKB is recommended over PKB.

Hygienic behaviour should be tested for at least twice during the season because results can vary due to different conditions at the time of testing. Keep in mind that a heavy nectar flow or feeding sugar syrup could stimulate hygienic behaviour; scientists typically test hygienic behaviour during a nectar flow or simulate a nectar flow by feeding sugar water.

Hygienic behaviour (UBO) can be tested with Unhealthy Brood Odour (UBeeO), which is a product sold by Optera. Instructions for how to test with UBeeO are available from the Optera website. UBO is recorded as the percentage uncapped two hours after spraying with UBeeO. The calculation is:

$$\frac{\textit{Number of uncapped cells at time 2} - \textit{Number of cells without capped brood at time 1}}{\textit{Number of cells treated} - \textit{Number of cells without capped brood at time 1}} \; x\ 100$$

A free guide for completing a hygienic behaviour test using liquid nitrogen is available (Frost 2014). If a 75 mm diameter tube is used, the approximate number of brood cells covered should be 100 (Büchler *et al.* 2024). Brood cells should contain pink- to purple-eyed pupae (Büchler *et al.* 2024). Hygienic behaviour (FKB) is recorded as the percentage of empty cells (i.e. dead brood removed) 24 hours after replacing the comb. The calculation is:

$$\frac{\textit{Number of empty cells at time 2} - \textit{Number of cells without brood at time 1}}{\textit{Number of cells treated} - \textit{Number of cells without brood at time 1}} \; x\ 100$$

Hygienic behaviour (PKB) is recorded as the percentage of unsealed cells six hours after replacing the comb (Büchler *et al.* 2024). Where 50 cells are treated, the calculation is: (50 – number of sealed cells after six hours) × 2.

Photo: Doug Somerville

Table 15. Suggested scores for traits on scale of 1 (low) to 5 (high).

Trait	1 (low/little)	2	3	4	5 (high/heaps)
Pollen	No frames	One frame	Two frames	Three frames	Four or more frames
Burr comb	0% of space filled	25% of space filled	50% of space filled	75% of space filled	100% of space filled
Propolis	0% of space covered	25% of space covered	50% of space covered	75% of space covered	100% of space covered
Aggression	Minimal smoke required, no aggression displayed	Requires more smoke as you progress	Requires more smoke, bees fly around you but do not attack	Stings, posturing, guards recruited	Large amount of smoke needed, multiple stings, persistent attack, bees chase you, sets off surrounding hives
Runny	Work as normal	A little movement	Some agitation and clustering	Rush about on combs and box, fairly clustered	Fly off and rush about on combs, bees grouped, leaving frames uncovered, no work being done
Colour (yellow, black, red, grey)	<70% of workers are the desired colour	70% of workers are the desired colour	80% of workers are the desired colour	90% or workers are the desired colour	All workers are the desired colour
Worker size	<13 mm	13 mm	14 mm	15 mm	>15 mm
Drone size	<15 mm	15 mm	16 mm	17 mm	>17 mm
Queen size	<19 mm	19 mm	20 mm	21 mm	>21 mm
Brood area	0-1 frames	1.25-3.25 frames	3.5-5.75 frames	6-7.75 frames	8 or more frames
Brood pattern/viability	<70%	71-80%	81-90%	91-95%	96-100%
Average pollen area in brood frames	0%	10%	20%	30%	>30%
Average honey area in brood frames	0%	10%	20%	30%	>30%
Swarming	No queen cells, no management required	Queen cells destroyed	Minimal management required – remove brood/honey/add super	Splitting was required	Hive swarmed despite management
Larval feeding of workers	Starving	Below-average feeding	Average feeding	Above-average feeding	Overfed
European foulbrood	No cells	1-5 cells	6-20 cells	21-30 cells	>30 cells
American foulbrood	Absent	Present			
Chalkbrood	0 cells	1-5 cells	6-40 cells	41-80 cells	>80 cells
Sacbrood	0 cells	1-5 cells	6-10 cells	11-20 cells	>20 cells
Nosema	Absent	Present			
Small hive beetle	0 beetles	1-5 beetles	6-40 beetles	41-80 beetles	>80 beetles
Wax moth	Absent	Present			
Hive cleanliness	Heavy layer of detritus	Unclean	Fairly clean	Very clean	Pristine

Table 16. Brood pattern/viability photo guide. Photos: Emily Noordyke.

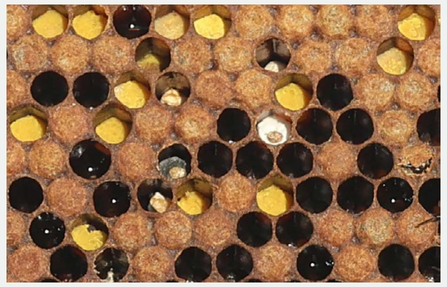
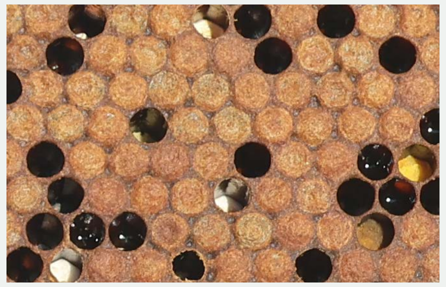
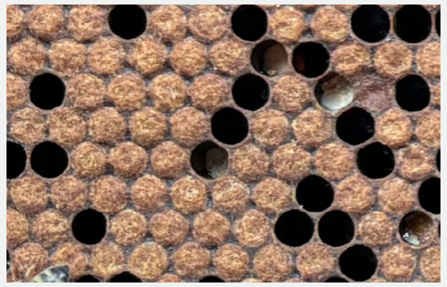
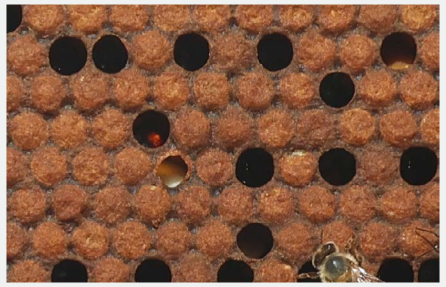
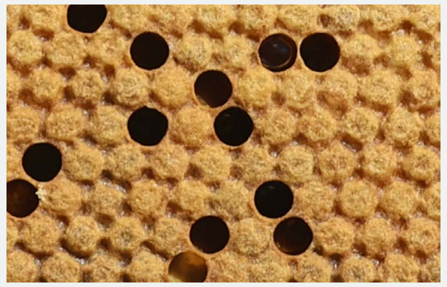
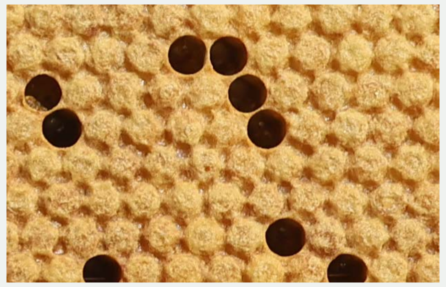
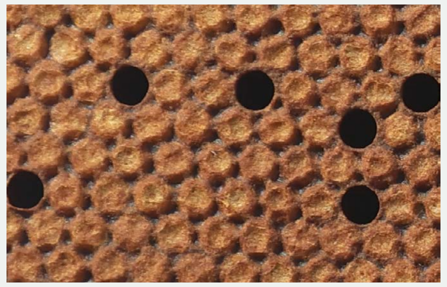
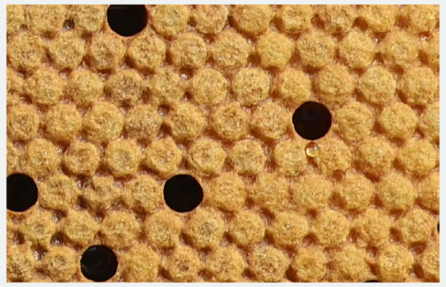
1: <70%		
2: 71-80%		
3: 81-90%		
4: 91-95%		
5: 96-100%		

Table 17. Average pollen area in brood frames photo guide. Photos: Emily Noordyke.

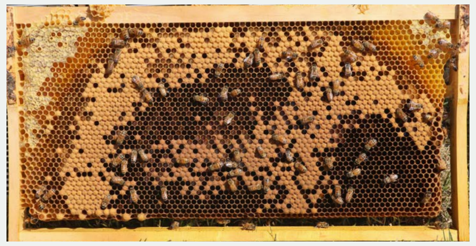
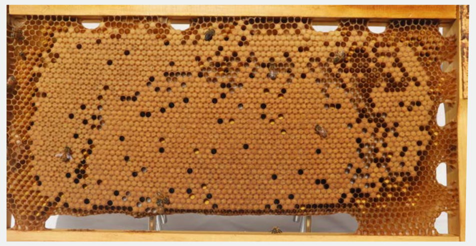
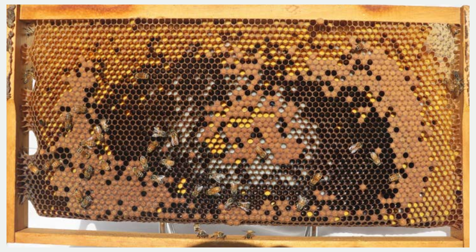
1: 0%	
2: 10%	
3: 20%	
4: 30%	
5: >30%	

Table 18. Average honey area in brood frames photo guide. Photos: Emily Noordyke and Nadine Chapman.

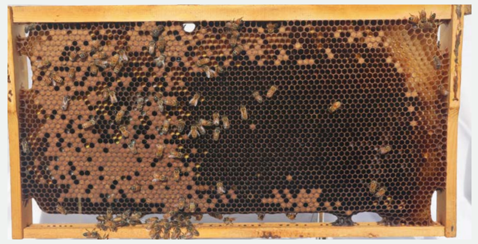
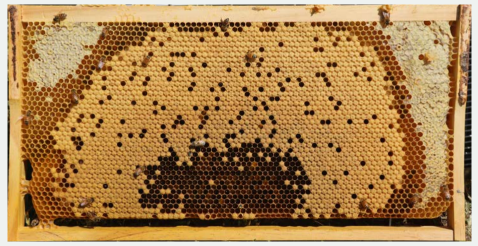
1: 0%	
2: 10%	
3: 20%	
4: 30%	
5: >30%	

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Photo: Kirra Hughes

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